Rock Mechanics and Rock Engineering (2022) 55:7629–7655
https://doi.org/10.1007/s00603-022-03059-4

ORIGINAL PAPER

# Determination of Mechanical Property Evolutions of Shales by Nanoindentation and High-Pressure $CO_2$ and Water Treatments: A Nano-to-Micron Scale Experimental Study

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Received: 21 January 2022 / Accepted: 29 August 2022 / Published online: 12 September 2022
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## Abstract

Fluid–shale interaction mechanisms and their implications on shale mechanical property evolutions are essential for both shale gas exploitation and $CO_2$ sequestration. To understand the potential influence of water and $CO_2$ exposure on shale, three shale samples from different formations were treated with dry supercritical $CO_2$, supercritical $CO_2$–water, and pure water for up to 30 days. Nanoindentation was used to measure the localized elastic modulus and hardness before and after fluid treatments. Combined nanoindentation with XRD–SEM–EDS analysis, we obtained the probing mechanical properties of three mineral phases in shales: quartz, organic matter, and mixed composites. The mechanical properties and creep behaviors of the mineral phases before and after fluid treatment were analyzed. XRD analysis and nanoindentation results reveal that the three shales are distinct in mineralogy and mechanical properties. The mechanical heterogeneity is related to the mineral constituents, which can be observed in the deconvolution results. The fluid treatment results indicate that dry supercritical $CO_2$ does not impair the nanomechanical properties of shale, whereas water-contained fluid shows a significant weakening effect. The mechanical evolution of shale is both time-dependent and mineralogy-related. After supercritical $CO_2$ and water treatment, the maximum deformations of three shales demonstrate an increasing trend, indicating a deterioration of the resistance to load. The elastic moduli and hardness of quartz, organic matter, and mixed composites exhibit apparent decreases after fluid treatment. The time-dependent mechanical behaviors of mineral phases were analyzed. $CO_2$ and water play critical roles in the reactions with shale and the involved mineral reactions are discussed. The experimental findings provide insight into the nanomechanical property evolutions of shale in field operations involved in $CO_2$ and water, such as fracturing and $CO_2$ sequestration.

## Highlights

- Shale mechanical property evolutions in fluid–shale interaction were traced on the nano-to-micron scale.
- Nanoindentation combined with XRD, SEM, and EDS was adopted to obtain the nanomechanical properties of mineral phases.
- The deformation of shales and creep behaviors of mineral phases were analyzed.
- The mechanical alteration after dry supercritical $CO_2$ exposure was inapparent while significant when water exists.

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**Keywords** Mechanical alteration · Shale · $CO_2$–rock interaction · Nanoindentation · Creep

## 1 Introduction

The fast growth of energy demand has forced unconventional energy resources, such as shale oil and tight gas into the spotlight (Warpinski et al. 2009; Barati and Liang 2014). Horizontal drilling and hydraulic fracturing are technologically applied in low-permeability reservoirs, to improve wellbore access, enhance the effective permeability, and finally achieve commercial energy flow (Barati and Liang 2014). In the exploitation process, excessive amounts of aqueous fracturing fluids have been injected into the target formations. These fracturing fluids contain muriatic acid, chemical modifiers, and gelling agents, which can not only deplete a good deal of water but also cause aquifer contamination (Osborn et al. 2011; Jackson et al. 2013). Thus, nonaqueous fracturing was proposed to reduce the environmental concerns and water-sensitive in high-clay content reservoirs (Middleton et al. 2015). As a nonaqueous fracturing fluid, $CO_2$ is favored for its nontoxicity and easy accessibility, as well as its use as an energized fluid in fracturing (Middleton et al. 2014). $CO_2$ has been gaining significant attention in recent years because of its miscibility with hydrocarbons, enhanced desorption of methane, and reduced pressurization requirements (Middleton et al. 2014, 2015). Injecting $CO_2$ into oil fields or gas fields for flooding or purging can not only improve recovery but also achieve geological storage of $CO_2$ (Rubin et al. 2015; Liu et al. 2017). Besides, it is an efficient and clean technology for shale gas extraction. In deep strata with high temperature and pressure, the injected $CO_2$ may transform from gaseous to a supercritical state ($ScCO_2$), which has very unique characteristics, such as high density, good solubility, fast gas diffusion, and low surface tension (Hu et al. 2016, 2017; Cai et al. 2019; Feng et al. 2019). The high density of supercritical $CO_2$ makes it a superior drilling fluid, and its low surface tension makes it easier to access fractures and pores. On one hand, the injected $CO_2$ could remain in the formation for months or even years (such as carbon geological storage). On the other hand, water introduction is unavoidable in horizontal drilling, and the flow-back water will be difficult to completely withdraw, causing water sorption and retention in the pore structures (Sang et al. 2019, 2020a, b). Besides, the shale reservoir may contain pre-existing water or brine. Therefore, the injected $CO_2$ is highly likely to dissolve in the water and create a weak acidic circumstance. This may lead to alteration in the transport and mechanical properties of the shale reservoir. In the absence of basic understanding, $CO_2$ injection's potential influences must be further explored, especially insights on the nano-to-micron scale.

$CO_2$–water–shale interaction is quite complicated because of the reversible ionization of carbon acid and the variation of solubility of $CO_2$ changes with temperature and pressure, which in turn dominates the physicochemical reaction between the mediums (Zhang et al. 2017). In previous studies, $CO_2$ and rock interaction caused obvious mechanical weakening of shale based on the mechanical tests of shale cores. It was observed that uniaxial compressive strength and Young's modulus of shale on a bulk scale reduce significantly after $CO_2$ treatment, where the reduction depends on the pressure, temperature, and time of the immersion (Lyu et al. 2016, 2018; Yin et al. 2017). Shale core samples with $CO_2$ adsorption go through distinct plastic phases before breaking down and show a lower degree of fragmentation, indicating a less brittle property than raw samples (Yin et al. 2017; Feng et al. 2019). Furthermore, porosity and permeability have been observed to be altered during $CO_2$ and rock interaction, which are two key parameters closely related to the transport properties of the porous media. Transport properties can either be enhanced or aggravated by dissolution or adsorption-induced swelling in $CO_2$ and rock interaction. The acidifying effect of $CO_2$ on the aqueous solution can dissolve organic matter, clay (montmorillonite, kaolinite), and carbonate (calcite) in shale, while silicates, such as quartz, are decomposed and precipitated into carbonate solids (Yin et al. 2016). These $CO_2$-induced reactions reshape the pore morphology and alter the permeability of reservoirs, transforming micropores (diameter, $d < 2$ nm) into mesopores ($d = 2$–$50$ nm) or macropores ($d > 50$ nm) (Yin et al. 2016; Ao et al. 2017; Pan et al. 2018; Sanguinito et al. 2018). However, the alteration caused by $CO_2$ is not always beneficial. Adsorption of $CO_2$ molecules on the shale surface reduces its surface potential energy, which results in swelling and an increase in surface area (Pan and Connell 2007). In some cases, $CO_2$ adsorption-induced swelling could cause volumetric strain, narrowing pores and fractures on the nanoscale, and inhibiting matrix permeability. (Pan and Connell 2007; Lu et al. 2016b; Wu et al. 2017). In another scenario, the swelling may be heterogeneous among the mineral components, which may give rise to new micro-crack generation in shale and improve the permeability (Yin et al. 2016). Both mechanisms may exist simultaneously, depending on the pressure and temperature at the formation, as well as the state of the injected carbon dioxide. For example, it is reported that the amount of swelling in coal caused by supercritical $CO_2$ was almost two times larger than that caused by

$CO_2$ in a sub-supercritical state (Perera et al. 2011). Further study is needed on the mechanism involved in the interaction between $CO_2$ and shale at the nanoscale (Huang et al. 2020).

Numerous studies have been conducted on the alterations in shale rock after exposure to $CO_2$ or its solution as aforementioned and found that the $CO_2$–water–rock interaction changes the mechanical strength of shale, as well as the pore structure, permeability, and mineralogy of the rock. Nevertheless, these measurements primarily pertain to shale alteration in bulk. For comprehensive insight into $CO_2$ flooding, sealing, and storage, it is essential to understand how the mineral's constituents respond to $CO_2$ exposure. Given the different mechanical and chemical properties of mineral constituents in shale, the influence of $CO_2$ on shale requires direct observation at the mineral level. Through the development of indentation instrument tests, such as nanoindentation, it is now possible to access the mechanical properties of rocks down to the nanoscale and expand the horizons of mechanical appraisal (Li and Bhushan 2002; Bobko and Ulm 2008; Shukla et al. 2013; Liu et al. 2016; Lu et al. 2020; Shi et al. 2020). In this study, a 30-day treatment of $CO_2$ and water was investigated to assess its effects on the major mineral constituents in shales. The nanoindentation test was used to identify mechanical changes before and after the treatment. By combining X-ray diffraction (XRD), scanning electron microscopy (SEM), and energy-dispersive X-ray spectroscopy (EDS), we identified the mineral phases below the indents and match the probing mechanical properties with corresponding minerals. This study offers direct mechanical measurements of the mineral constituents in the treated shale samples and provides an insight into the $CO_2$–water–rock interaction from a microscopic perspective.

## 2 Experimental Methodology

### 2.1 Sample Description and Preparation

Three shale samples were used in this study. Shale samples M-1 and M-2 were collected from the Allegheny formation in Pennsylvania. Shale sample I-1 was recovered from a drill core from the Illinois basin. The shale sampling locations are shown in Fig. 1. Due to their various mineral compositions and distinct geological characteristics, these samples provide a variety of mechanical properties of specific minerals in nanoindentation and increase statistical reliability.

The shale cores were first cut into cubes, as shown in Fig. 2a. The residual chips were pulverized down to 325 mesh powder (equivalent to 44 μm) for XRD analysis. All the sample cubes and powder were oven-dried for 24 h at 100 °C. Following that, all sample cubes prepared for nanoindentation tests were in epoxy resin to prevent them from spalling during testing, as shown in Fig. 2b. A flat surface is necessary for nanoindentation tests to guarantee accuracy and consistency (Donnelly et al. 2006; Miller et al. 2007). Thus, fine polishing was conducted on the MultiPrep Surface Preparation Machine (Allied High-Tech Products, Inc.). After the epoxy resin had been fully cured, the sample

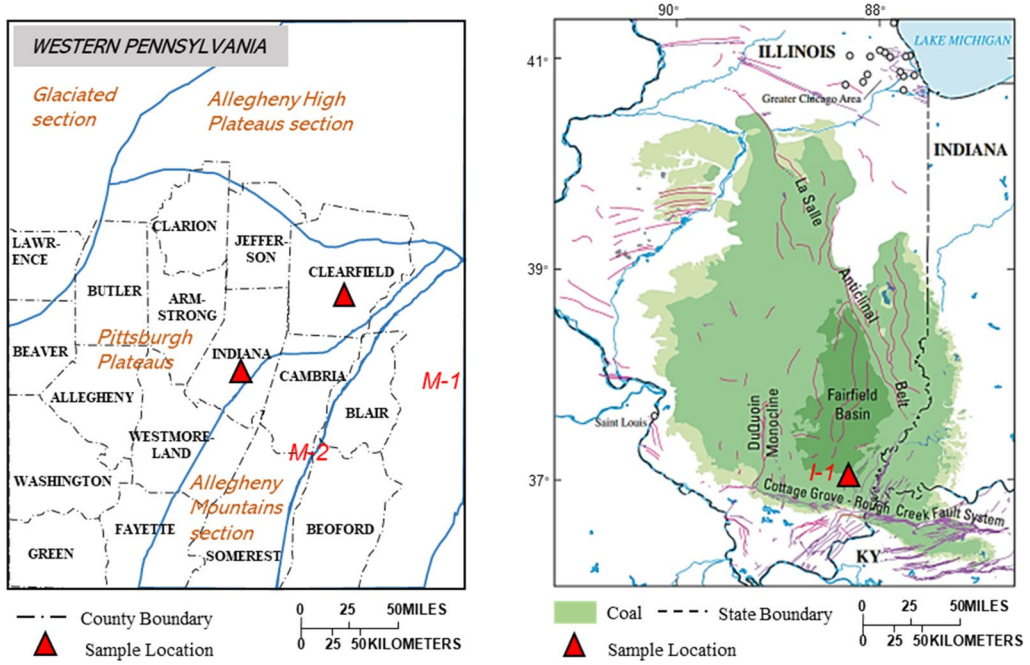



**Fig. 1** Geological location of the tested shale samples (Ferm 1955; Puglio and Iannacchione 1979; Hatch et al. 2002)

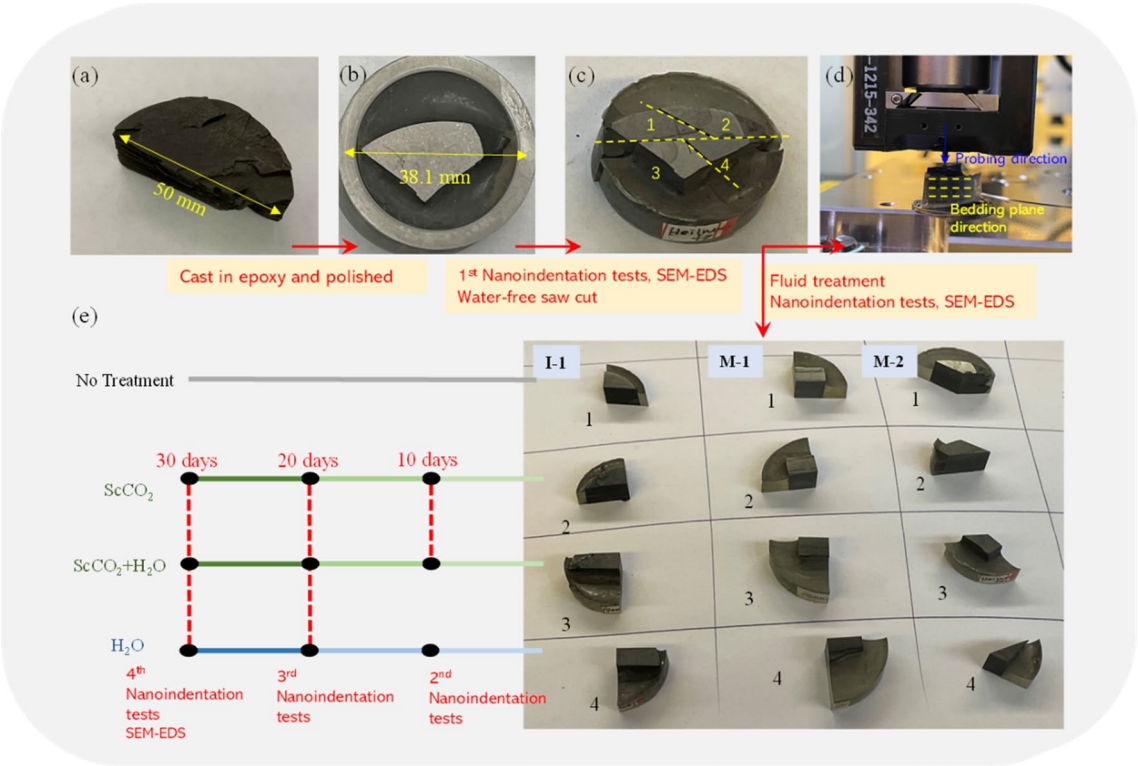



**Fig. 2** Sample preparations for nanoindentation tests and fluid treatment experiments

was attached to a metal disk with a crystal bond for ease of polishing. The shale cubes underwent a precise polishing procedure using abrasive papers from P1200 to P4000. At this stage, ethanol and lubricants with alcohol were used to avoid mineral hydration and smooth the process. To reach a finer surface, the samples were further polished with 1 µm polycrystalline diamond suspension, 0.25 µm polycrystalline diamond suspension, and 0.05 µm alumina suspension. All suspensions were ethylene- or glycol-based to avoid potential mineral hydrations. After the polishing process was completed, the surfaces of the samples were cleaned with ethanol, and then dried with pressurized filtered air. The surfaces that are being polished are all parallel to the bedding plane direction. An optical profilometer was used to check the roughness of the polished surfaces.

## 2.2 Nanoindentation Tests and SEM–EDS Analysis

The experiments were conducted in three steps: (1) nanoindentation tests and SEM–EDS analysis before treatment on raw shale specimens; (2) Supercritical $CO_2$ and water treatments under different conditions; (3) Nanoindentation tests, SEM and EDS analyses on the treated specimens. As shown in Fig. 2, the first round of nanoindentation tests and SEM–EDS analysis were performed on the polished shale samples to obtain the mechanical properties before fluid treatments. Then, the samples were carefully cut into four segments using a water-free saw, marked as "1", "2", "3", and "4" in Fig. 2c, e. No fluid treatment was conducted on segment "1". Segments "2" were treated with supercritical $CO_2$ for 10, 20, and 30 days. Segments "3" were treated with supercritical $CO_2$ and water for 10, 20, and 30 days, while segments "4" were treated with water for 20, and 30 days. After 10-day and 20-day fluid treatment, the second round and third round nanoindentation tests were conducted on the samples. After the 30-day treatment, the fourth round nanoindentation of tests was conducted, followed by SEM–EDS analysis. The probing direction in nanoindentation tests was vertical to the bedding plane direction, as shown in Fig. 2d.

Nanoindentation tests were conducted on the Nanoindenter at Penn State Material Characterization Lab. The apparatus is shown in Fig. 3a, and the technical specifications of the apparatus can be found in the previous study (Liu et al. 2021). SEM–EDS analysis was conducted on the Zeiss Merlin SEM and Oxford EDS detector. The grid indentation method is commonly used to analyze the mechanical properties via nanoindentation (Liu et al. 2018a). Before nanoindentation tests, cross marks were engraved on the center of shale samples, which divided the surface into four regions. A grid indentation with $8 \times 8$ was conducted in each region, as shown in Fig. 3b.

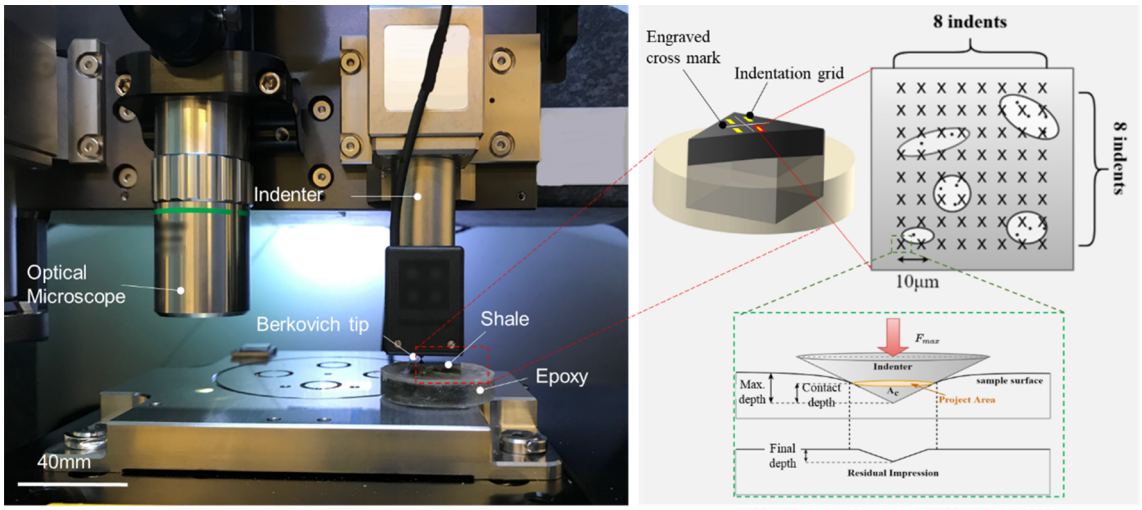



**Fig. 3** **a** Nanoindenter and the polished shale sample; **b** schematic of nanoindentation tests

The distance between two adjacent indents was 10 μm. A maximum load of 10 mN was used in the nanoindentation tests. A typical loading function that contains a loading process, a holding period, and an unloading process was applied, and the durations were 0.5 s, 0.2 s, and 0.5 s, respectively. The loading function was controlled by the accelerated property mapping mode. Through the coupling of a high-bandwidth electrostatically actuated transducer, a fast control system, as well as data acquisition electronics, the accelerated property mapping mode is capable of measuring nanomechanical properties at six measurements per second. It can quickly screen and locate surface features and spatial distributions of shale properties using optical and scanning probe microscopy. Specifically, the cross marks are intended to aid in positioning the indentation grid for subsequent SEM–EDS analysis. Initially, the cross mark can be easily found by the optical microscope in the Nanoindenter. Then, a site near the mark that was relatively flat was picked for nanoindentation tests. The coordinates of the indentation grid relative to the cross mark can be obtained by the optical microscope's scale ruler. With the help of these coordinates, indentation grids on shale surfaces can be more easily relocated when SEM–EDS analysis was performed. Previous numerical

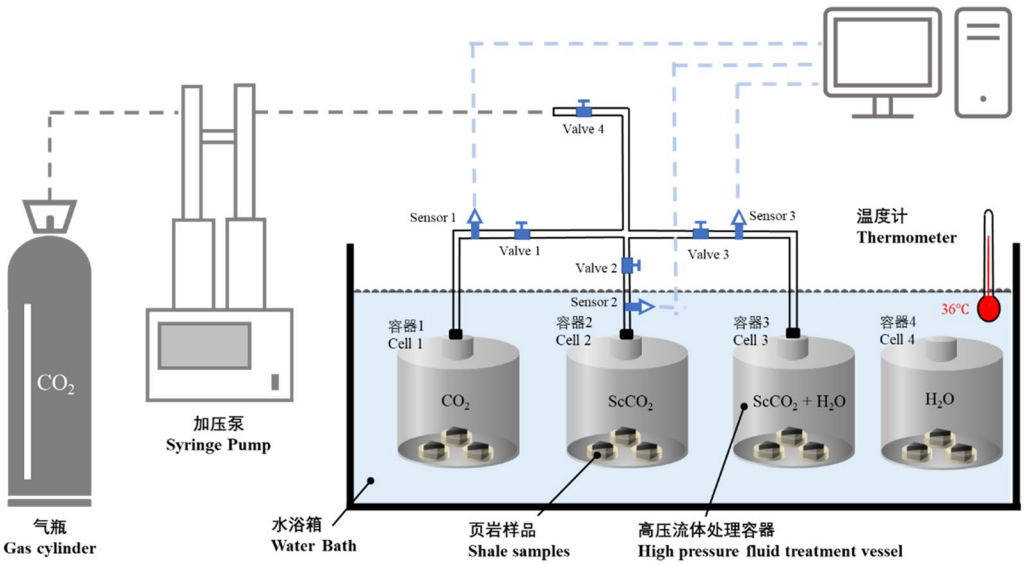



**Fig. 4** Schematic of the $CO_2$ and water treatment process on shale samples

simulations of scratch tests showed that local stress concentrations caused by scratches could affect stress distributions in areas equal to the scratch width (Randall et al. 2009; Varga et al. 2017). To prevent additional stress caused by the engraving, the distances between the indentation grids and cross marks were at least 2–3 times the width of the cross marks.

### 2.3 $CO_2$ and Water Treatment Process

Figure 4 shows the schematic of the $CO_2$ and water treatment process on shale samples. The shale segments were put into vacuumed high-pressure fluid treatment vessels. In vessel 1, dry $CO_2$ was injected under the pressure of 1500 psi through a syringe pump. In vessel 2, High Performance Liquid Chromatography (HPLC) water was added to it and then supercritical $CO_2$ was injected under the pressure of 1500 psi. In vessel 3, only HPLC water was added. Each vessel was filled with a segment of M-1, M-2, or I-1 shale, as shown in Fig. 4. A thermostatic water bath was used to maintain the temperature of the whole treatment system at 36 °C, which is over the critical temperature of supercritical $CO_2$ at 31.04 °C. The pressures in vessel 1 and vessel 2 were monitored by pressure sensors to ensure that the pressure was over the critical pressure of $CO_2$, so that $CO_2$ stayed in a supercritical state during the treatment. Previous studies showed that the adsorption equilibrium between $CO_2$ and the shale samples was reached after 8–10 days (Yin et al. 2016, 2017). Thus, the treatment durations were set at 10, 20, and 30 days to study the time-dependent effects associated with $CO_2$–water–shale interactions. Following each treatment period, the samples were removed and nanoindentation tests were conducted within 6 h to minimize exposure to air.

The workflow of the experiments is shown in Fig. 5. XRD analysis gives the mineralogy information of the shale samples. Nanoindentation tests characterize the mechanical properties of shales before and after treatment for tracing mechanical property alteration. SEM assists to locate the indents and offers the morphology of the residual impressions, whereas EDS provides information regarding the chemical composition of the indented surface and facilitates the identification of minerals. Combining the XRD–SEM–EDS results with nanoindentation, the mineralogy-related mechanical change can be directly observed.

## 3 Experimental Results

### 3.1 Mineralogy, Morphology, and Chemical Characterization of Shales

Table 1 summarizes the mineralogical compositions of the shale samples, determined by XRD analysis. Sample M-1 is primarily composed of silicates (quartz, muscovite) and clay minerals (kaolinite, chlorite), which together account for a weight fraction of 99%. A small amount of pyrite and rutile were also detected in M-1, but no carbonates or organic material were observed. Sample M-2 is mainly comprised of quartz, diopside, clay minerals (mainly illite, kaolinite, and chlorite), siderite, rutile, pyrite, as well as 15% organic matter. The total organic carbon (TOC) of shale M-2 range from 1.5 to 6.2 wt% (Jarboe et al. 2015). Sample I-1 is mainly composed of silicates (quartz, muscovite, and albite), clay minerals (chlorite, kaolinite, and illite), carbonates (dolomite, calcite), and pyrites, as well as a considerable abundance of organic matter (37.4%). The TOC content of shale I-1 is 21.4 wt% (Sang et al. 2019).

Figure 6 shows the surface morphology and chemical composition characterizations of the shale samples, analyzed by SEM–EDS after the first round of nanoindentation tests. SEM images enable an intuitive understanding of the experimental scale of nanoindentation testing. Also displayed are the distribution of the residual impressions and the surface morphology of the shales, ready for further analysis. The probing residual indents in the dotted white circles can be observed in SEM images. EDS analysis shows that silica, oxygen, and aluminum are the most abundant elements in

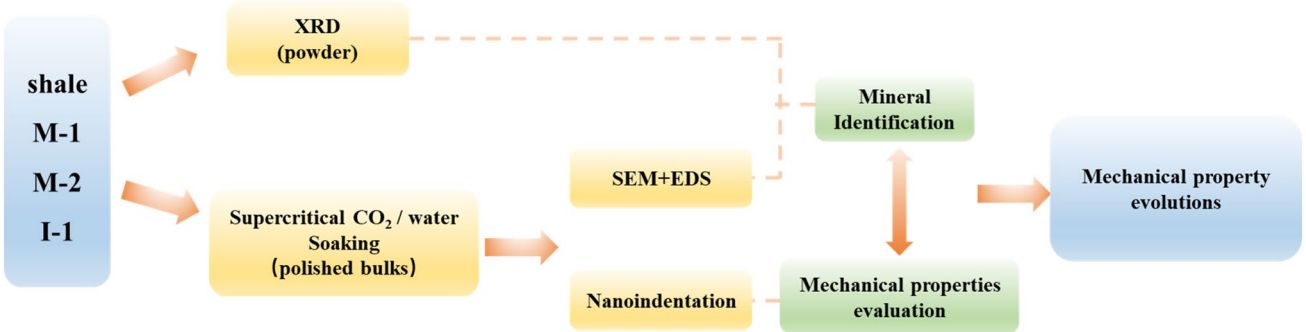



**Fig. 5** Illustration of the experimental methods

**Table 1** Weight percentage of the mineralogical compositions in shale samples (wt%)

| Samples | Silicates | | | | Amorphous | Clay | | | | Sulfides | Carbonates | | | Other |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Quartz | Muscovite | Albite | Diopside | Organic matter | Kaolinite | Clinochlore | Chlorite | Illite | Pyrite | Siderite | Dolomite | Calcite | Rutile |
| M-1 | 32.6 | 45.5 | – | – | – | 11.3 | 9.5 | – | – | 0.3 | – | – | – | 0.7 |
| M-2 | 23.4 | – | – | 6.0 | 15.0 | 10.9 | 9.1 | – | 18.0 | 0.3 | 8.2 | – | – | 7.8 |
| I-1 | 11.7 | 17.0 | 2.5 | – | 37.4 | 1.6 | – | 6.6 | 17.4 | 1.6 | – | 3.1 | 0.9 | – |

shale M-1 and M-2, while carbon, oxygen, and silica are the most abundant elements in I-1. As is shown from the SEM image in Fig. 6a, shale M-1 is characterized by a large number of strip-like layered silicates on its surface. Coarse-sized silica-rich particles are observed, i.e., the magenta blocks in the EDS image, which are inferred to be quartz. In Fig. 6b, it is observed that shale M-2 contains relatively rich iron-bearing minerals and blocky organic compounds. A closer look at the coarse grain size of shale M-2 shows it to be smaller than the coarse grain size of shale M-1 sample, and the distribution is more fragmented. There is plenty of carbonaceous material and the least coarse-grained minerals in shale I-1, as shown in Fig. 6c. Clearly, I-1 shale is organic-rich, which is consistent with the peak of the carbon element in the EDS spectrum and the amorphous component in the XRD results. Although XRD showed no occurrence of organic matter in shale M-1, we found trace organic matter in the sample by SEM–EDS, as shown in Fig. 6a.

Combining the SEM and EDS elemental analyses, the main minerals can be identified and recognized. For large particles, the mineral can be identified by the abundance of the characteristic chemical elements. For example, the particles that show extremely rich in silica, while the absence of aluminum is quartz, and the particle abundant in iron and sulfur is pyrite. For the minerals that cannot distinguish at the first sight, local EDS analysis can help. Local EDS analysis provides element quantification within the selected boundary of the target mineral. Combining the local element composition and the XRD result, the target mineral can be identified. Figure 7 shows several local element spectrums on the specific particles, which can be identified as quartz, pyrite, organic matter, and rutile. The region-specific mineral distribution and elemental composition of the shales can provide direct reasoning for the mineralogy-related mechanical data in nanoindentation tests.

## 3.2 Initial Mechanical Properties of Shales

The load–displacement curves are recorded during the nanoindentation tests, which monitored the load and the penetration depth of the probe and can be used to determine the elastic moduli and hardness of the samples (Fischer-Cripps and Nicholson 2004). Elastic modulus is determined by the initial slope of the unloading curve and the elastic parameters of the indenter tip. Hardness is calculated by dividing the indentation load by the projected area, which can be indirectly determined by the penetration depth and geometry of the indenter. The detailed derivation of the elastic modulus and hardness in a nanoindentation test can be found in previous studies (Oliver and Pharr 1992; Pharr and Oliver 1992). Load–displacement curves can not only reflect the mechanical properties of the tested material but also reveal

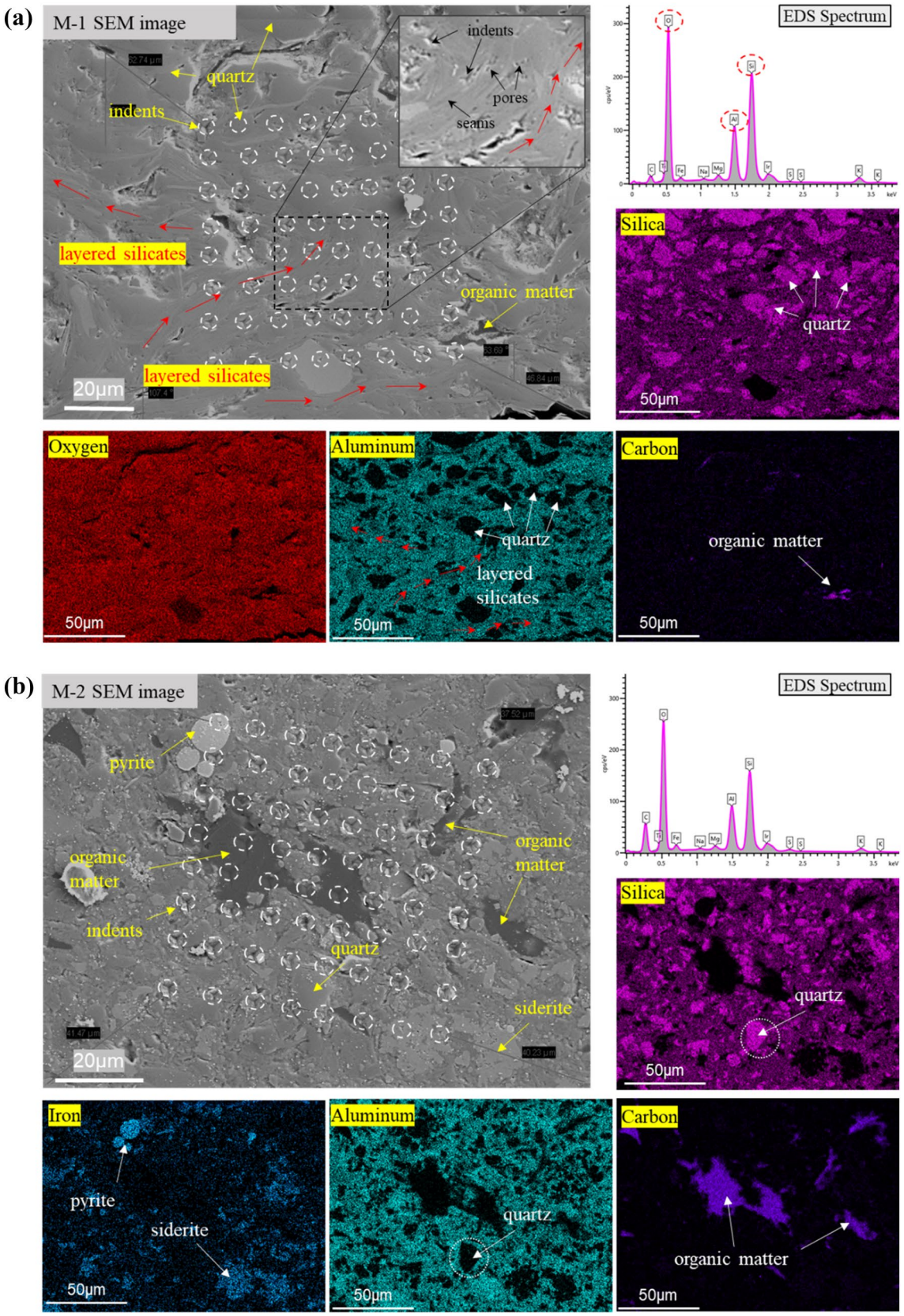



**Fig. 6** Morphology and chemical composition characterization of the shale samples by SEM–EDS analysis. **a** Shale M-1; **b** Shale M-2; **c** Shale I-1

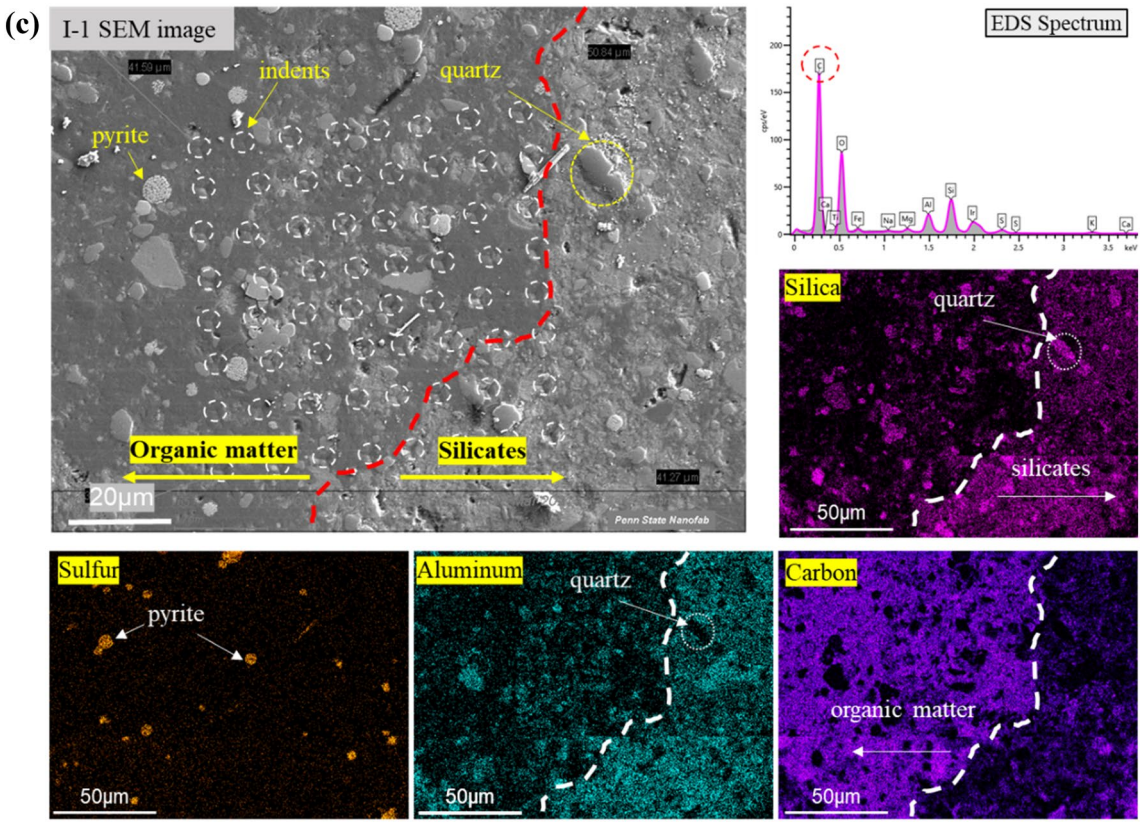



**Fig. 6** (continued)

the microstructure and specific events, such as phase transformation, atoms dislocation, and cracking (Domnich et al. 2000; Fischer-Cripps and Nicholson 2004; Liu et al. 2021). Figure 8 shows the load–displacement curves of the three shale samples before fluid treatments. Different maximal penetration depths indicate different deformation-resistances to load among the samples. The majority of the maximum depths of shale M-1, M-2, and I-1 range in 200–600 nm, 200–1200 nm, and 800–1000 nm, respectively. The differences in maximum depth are attributed to the mineralogical diversity of the shale samples. The penetration depth depends on the hardness of the mineral beneath the probe. Accordingly, shale containing more hard minerals such as quartz and pyrite presents less deformation, while shale containing many soft substances such as organic matter has a large penetration depth. Scattering curves for M-1 shale and M-2 shale demonstrate high heterogeneity, whereas curves for I-1 show less heterogeneity due to the abundant organic matter distributed widely in the tested area.

Figure 9 shows the frequency distributions of elastic moduli extracted from load–displacement curves. In shale M-1, elastic moduli range from 21.4 to 104.0 GPa, whereas in shale M-2, elastic moduli range from 7.6 to 125.6 GPa. The elastic moduli of shale I-1 range from 3.5 to 38.6 GPa. The elastic moduli obtained from nanoindentation tests are local mechanical measures of the mineralogical phases beneath the probe tip. Mineral constituents concerning their mechanical properties are commonly separated using the deconvolution method (Cała et al. 2017; Veytskin et al. 2017; Li et al. 2019; Manjunath and Jha 2019; Graham et al. 2020; Luo et al. 2020). To produce a better display, a Gaussian mixture model (GMM) is used to describe the mineral phases in shales. The mechanical properties of each phase are investigated using a statistical cluster deconvolution technique under the assumption that the properties fit a normal distribution. Probability density functions (PDF) are fitted to the experimental frequency plots of the elastic moduli. Three mechanical phases are identified in shale M-1: cluster 1, cluster 2, and cluster 3. The mean values of the clusters are 40.3 GPa, 54.0 GPa, and 72.2 GPa, respectively. The mean values represent the mean phase properties of the soft minerals, minerals with mediated stiffness, and hard minerals in shale, which correspond to clay minerals, muscovite/mineral composite, and quartz/iron-bearing minerals, respectively. Two phases are distinguished in shale M-2, with mean values of the clusters 30.8 GPa and 54.4 GPa, which refer to the mechanical properties of soft minerals such as organic matter/clay minerals and hard minerals, such as quartz. The

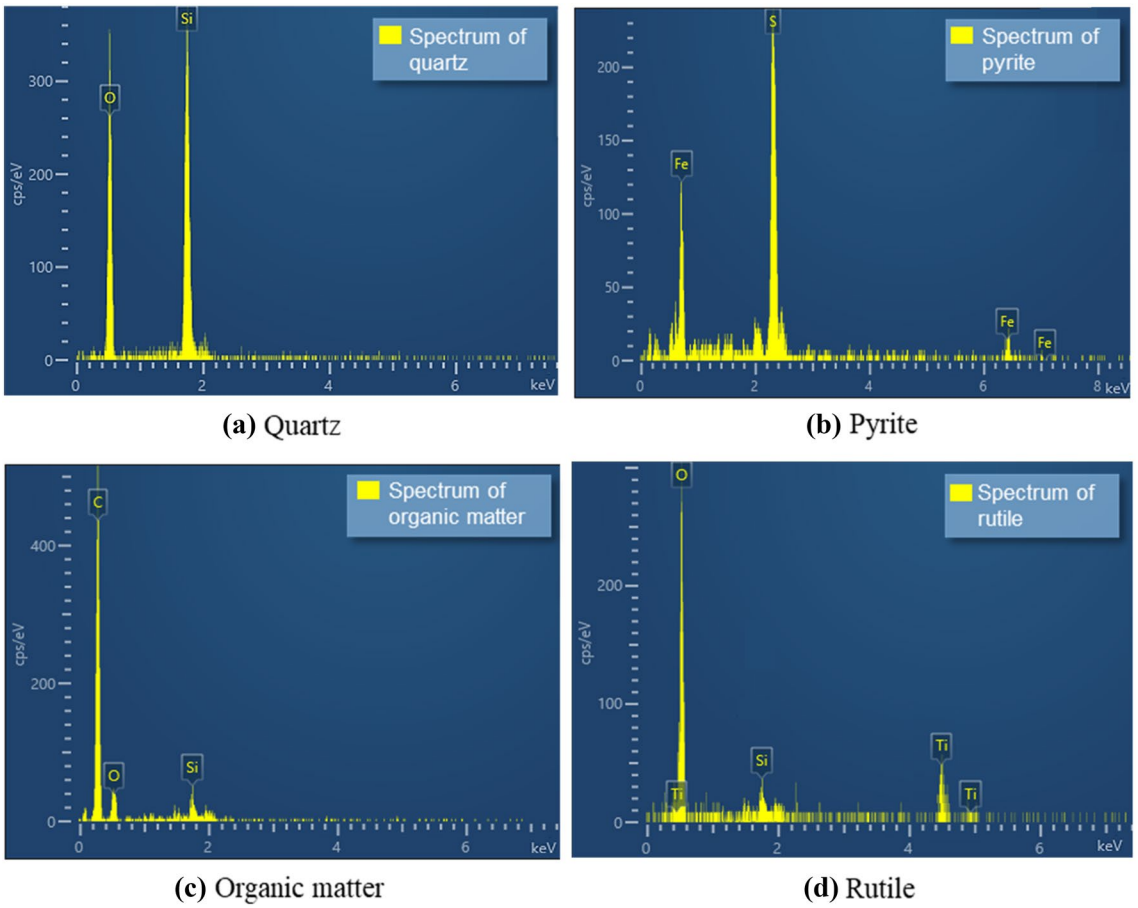



**Fig. 7** Local element spectrums on the specific particles by EDS analysis. **a** Quartz; **b** pyrite; **c** organic matter; **d** rutile

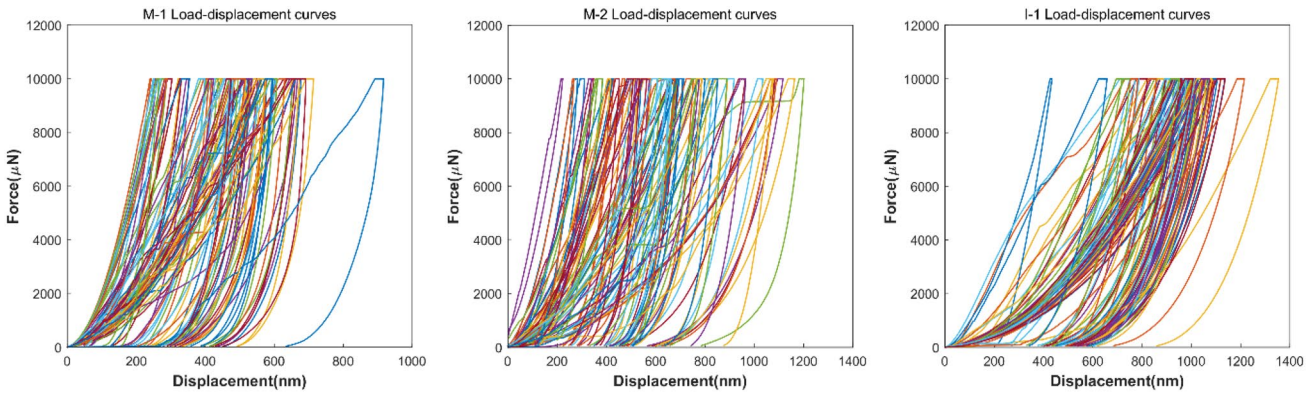



**Fig. 8** Load–displacement curves of M-1, M-2, and I-1 shales before fluid treatments

mean values of the two clusters identified in shale I-1 are 8.9 GPa and 15.8 GPa, which are significantly softer than other shales owing to the presence of mass organic matter.

An interesting observation is the apparent variation in elastic modulus of the same mineral between different shales. shale I-1 exhibits elastic moduli ranging from 10.3 to 38.6 GPa, which are considerably lower than the measured elastic moduli for shale M-1 and M-2. Previous studies have also observed this phenomenon as a 'substrate effect' (Doerner and Nix 1986; Saha and Nix 2002; Yang et al. 2020). Several granular mineral clusters can be seen on the surface of shale I-1, such as quartz and pyrite, which

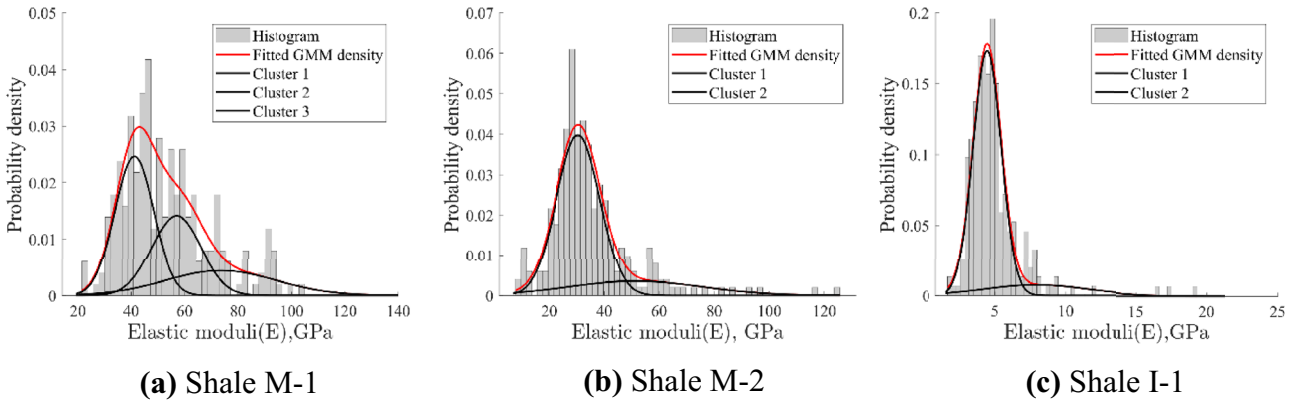



**Fig. 9** Frequency distributions and deconvolution results of the elastic moduli of shale samples

are isolated from each other and enclosed in carbonaceous material, as shown in Fig. 6c. When the indentation load is applied to the grains, organic matter around and underneath the grains plays a role of a soft substrate. As illustrated in Fig. 10, the deformation of the soft substrate yields an overestimation of the penetration depth, which leads to an underestimation of hardness. It also causes a weak resistance to elastic deformation and thus results in a lower measurement elastic modulus. Similarly, the elastic modulus of organic matter shows a large discrepancy in shale M-2 and I-1, which may be attributed to the different stiffness of the substrate between shales. In addition, the mechanical property of organic matter is related to thermal maturity (Abedi et al. 2016). The vitrinite reflectance ($\%R_0$) value of Shale M-2 is 1.3 to over 2.0, which is a high thermal maturity. The vitrinite reflectance ($\%R_0$) value of Shale I-1 is 0.71, indicating a low maturity. A previous study found a decrease in aliphatic C–H bonds and an increase in aromatic compounds

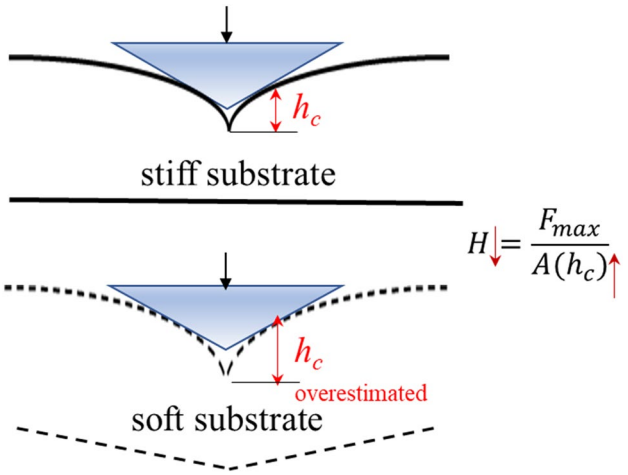



**Fig. 10** Schematic of the underestimation of hardness due to the overestimation of contact depth of indenter when probing on soft substrate

occur when kerogen matures, which increases the density of kerogen and leads to a stiffer property (Zhao et al. 2020). Overall, comparisons of probing mechanical properties between different shales should be made with great caution. The substrate effect should be considered when interpreting the mechanical properties of different shale types.

## 3.3 Mechanical Alterations After Supercritical $CO_2$ and Water Treatments

### 3.3.1 Load–Displacement Curves

To compare the load–displacement of shale before and after fluid treatments, we extract the mean load–displacement curve of each experimental condition. 256 load–displacement curves were generated in each experiment group, which are dispersive due to the heterogeneity of shale, as illustrated in the black curves in Fig. 11a. A mean load–displacement curve can be calculated based on all the load–displacement curves in the group, as shown in the red curve in Fig. 11a. The mean load–displacement curve provides a statistical representation of the mechanical response under specific experimental conditions. Figure 11b–d displays the mean load–displacement curves of three shales before and after treatment with supercritical $CO_2$ and water, shown in red curves and blue ones, respectively. Regardless of the treatment condition, the curves of each sample are quite similar. In each sample, the shapes of the curves are quite similar regardless of the treatment condition. For shale M-1, the curve after treatment with supercritical $CO_2$ shows less penetration depth than the initial one, while for shale M-2 and shale I-1, the curves after fluid treatment are almost the same as the initial ones. In all the shales, the curves after supercritical $CO_2$–water and pure water treatment show larger deformations than the initial ones, indicating a mechanical weakening after fluid treatment. After pure water treatment, the curves demonstrate the most prominent deformation in

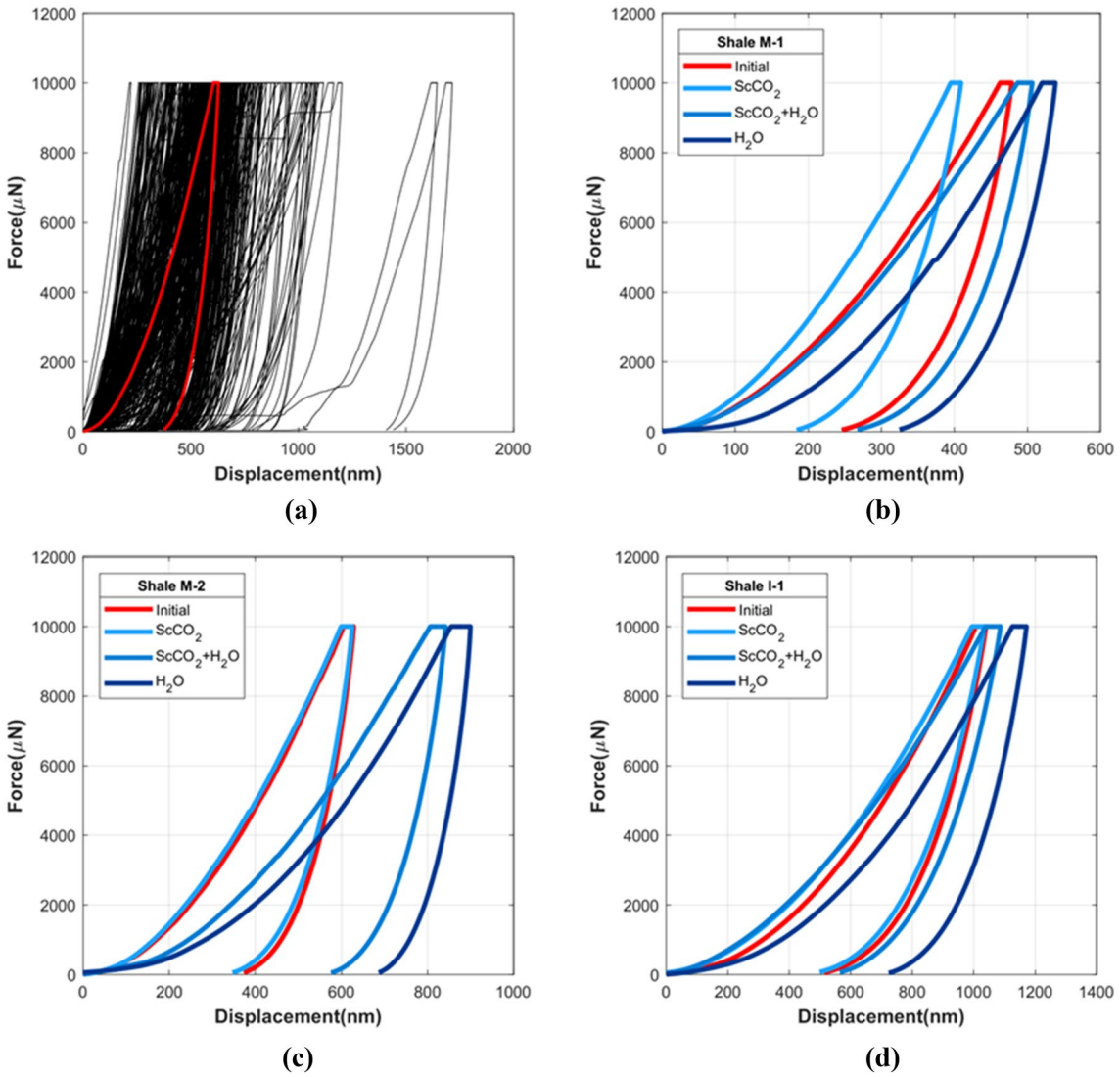


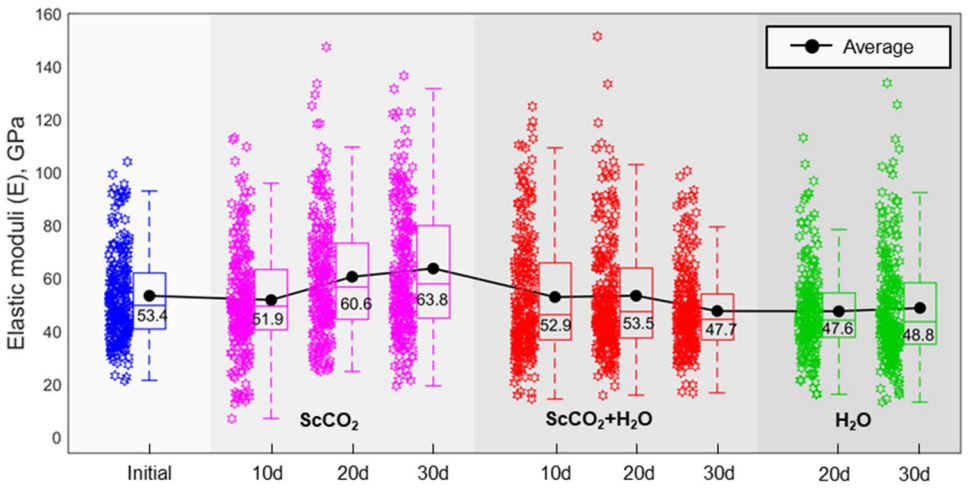

**Fig. 11** Load–displacement curves of shales. **a** Schematic of the mean load–displacement curve; **b** mean load–displacement curves of shale M-1 before and after fluid treatment; **c** mean load–displacement curves of shale M-2 before and after fluid treatment; **d** mean load–displacement curves of shale I-1 before and after fluid treatment

the loading, holding, and unloading processes, which suggests a significant mechanical alteration of the shales.

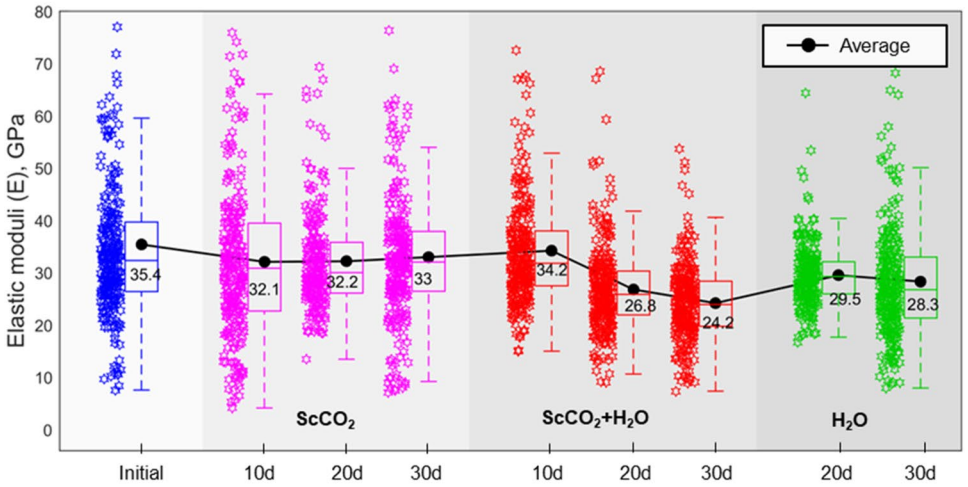
### 3.3.2 Elastic Modulus and Hardness

The elastic modulus and hardness of shale samples after fluid treatment are shown in Figs. 12 and 13. Both scatter plots and box charts are used to visualize the distribution of the mechanical parameter values. Box charts can easily depict discrete distributions of data and are less prone to outliers. In the box chart, the line below the box represents the upper limit, and the line above the box represents the lower limit. Black dots indicate average values. The bottom, center, and top lines of the box indicate the first quartile, second quartile (median), and third quartile values of the group, respectively. To reduce the uncertainty caused by the different substrates mentioned above, the probing results are compared between the same shale.

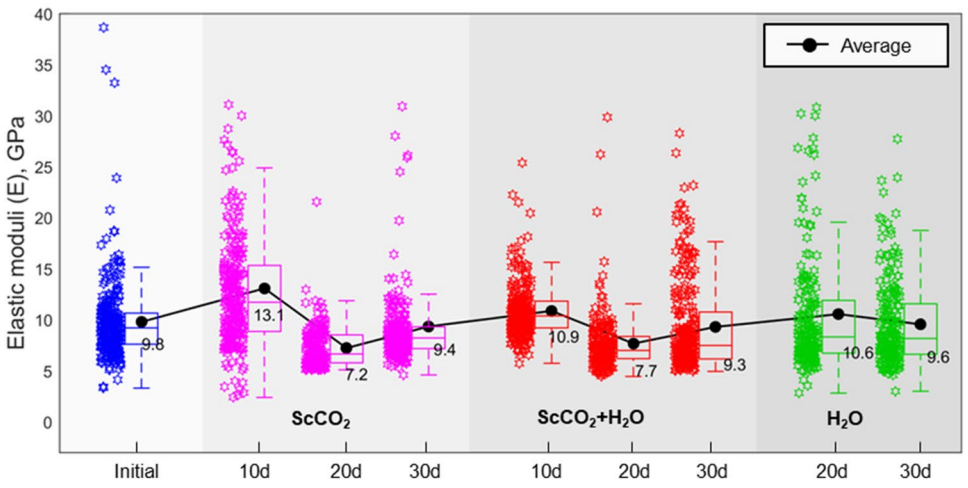
The elastic moduli of shale samples after different fluid treatments are shown in Fig. 12. Without any fluid treatment, the initial average elastic moduli of M-1, M-2, and I-1 shale are 53.4 GPa, 35.4 GPa, and 9.8 GPa. After dry supercritical $CO_2$ treatment, the elastic moduli of shale M-2 decrease, while M-1 and I-1 shale are a little higher than those of untreated samples. The reason for the increase is that the mineral distribution within the indentation area varies greatly. The average value of the probed data will rise as the indenter lands on more high-strength minerals. After 30 days of treatment with the water-contained fluid (i.e., supercritical $CO_2$–water and pure water), the elastic




(a) Shale M-1




(b) Shale M-2




(c) Shale I-1

**Fig. 12** Alterations on the elastic modulus of shales under different treatment conditions. **a** Shale M-1; **b** Shale M-2; **c** Shale I-1

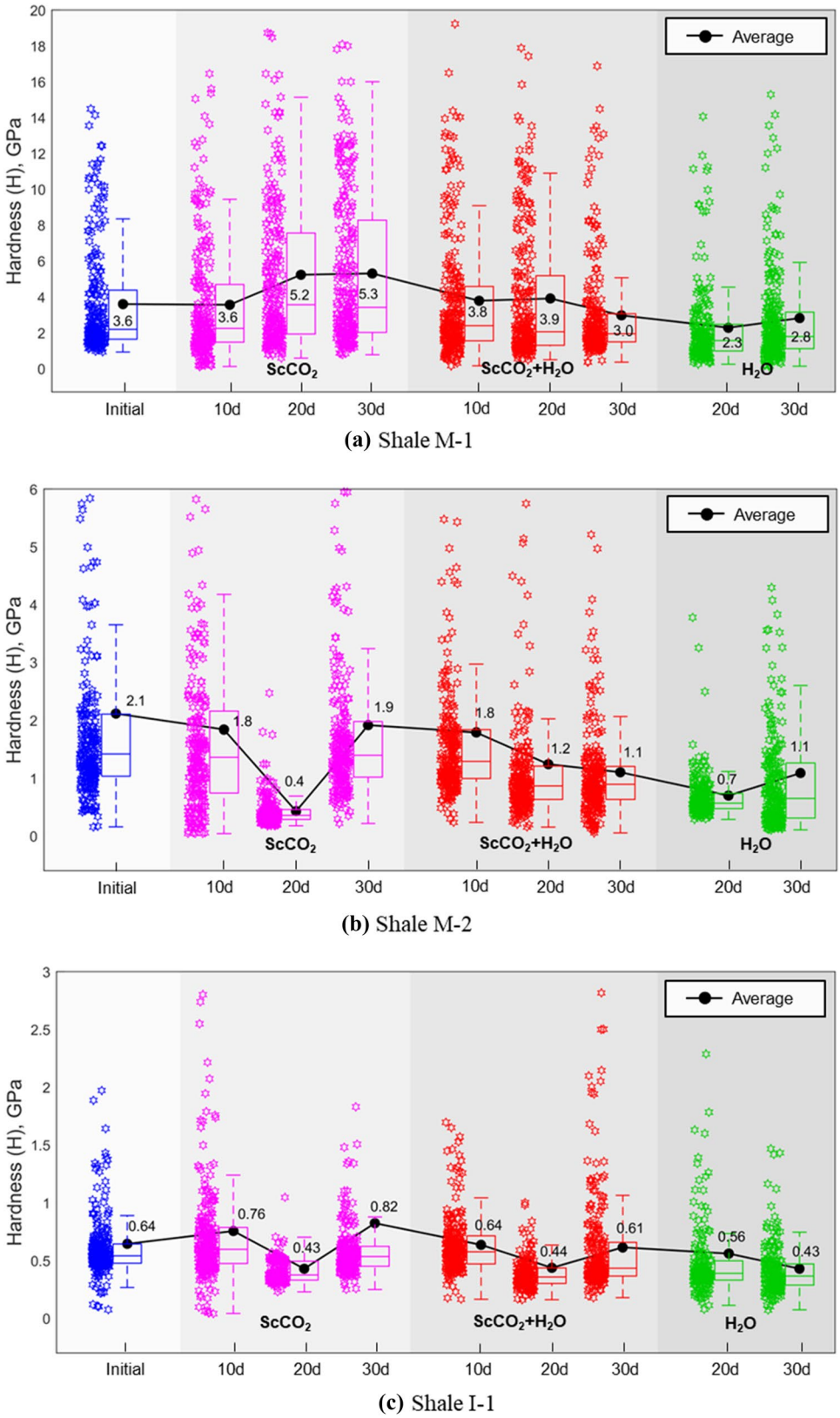



**Fig. 13** Alterations in hardness of shales under different treatment conditions. **a** Shale M-1; **b** Shale M-2; **c** Shale I-1

**Table 2** Mechanical reduction of the shales after 30 days of water-contained fluid treatment

| Samples | Elastic modulus ($E$) | | Hardness ($H$) | |
|---|---|---|---|---|
| | $ScCO_2$ and water (%) | Water (%) | $ScCO_2$ and water (%) | Water (%) |
| M-1 | 10.7 | 8.6 | 16.7 | 22.2 |
| M-2 | 31.6 | 20.1 | 47.6 | 47.6 |
| I-1 | 5.1 | 2.0 | 4.7 | 32.8 |

moduli of the three shale samples all witness a decrease to varying degrees, especially in M-2 shale. The average elastic modulus of M-1, M-2, and I-1 shale decreased by up to 10.7% (30 days), 31.6% (30 days), and 21.4% (20 days) after treatment with supercritical $CO_2$ and water. After 30 days of pure water treatment, the average elastic modulus of M-1, M-2, and I-1 shale decreased by 8.6%, 20.1%, and 2.0%, respectively, as shown in Table 2.

Figure 13 illustrates the alteration in the hardness of shale samples after different fluid treatments. After dry supercritical $CO_2$ treatment, the hardness of shale M-1 and I-1 increases, whereas the hardness of shale M-2 declines. Particularly, M-2 has a lower hardness after 20 days of treatment with supercritical $CO_2$ and has a more concentrated distribution than the other groups. This is due to the grid nanoindentation landing on relatively soft organic matter on the shale surface. As shown in Fig. 6b, organic matter is observed in blocks on the surface of shale M-2. The hardness can be very low and homogeneous when the indentation grid falls on soft organic matter. Following 30 days of treatment with the water-contained fluid (i.e., supercritical $CO_2$–water and pure water), the indentation hardness of the three shale samples declines to varying degrees, as shown in Table 2. After supercritical $CO_2$–water treatment, the mean hardness of M-1, M-2, and I-1 shale decreased by 16.7% (30 days), 47.6% (30 days), and 31.3% (20 days), respectively. After 30 days of pure water treatment, the mean elastic modulus of M-1, M-2, and I-1 decreased by 22.2%, 47.6%, and 32.8%, respectively. The different alterations in elastic modulus hardness of shales are related to the different mineral compositions, which demonstrate a variety of reactions with $CO_2$ and water. The involved mineral dissolution and precipitation are discussed in detail in the discussion.

The evolution of probed mechanical properties of shale samples after treatments is depicted on identical axes for comparison, as shown in Fig. 14. The overall mechanical strength of the shales decreases as the amount of organic phase (i.e., M-1 < M-2 < I-1), while the degree of mechanical homogeneity of shale increases. This manifests as the scattered data points in shale M-1 and overlapping data that span a narrow range in shale I-1. For shale M-1, with the increase of treatment duration, the mechanical properties after supercritical $CO_2$ treatment become slightly scattered, while that of supercritical $CO_2$–water treatment showed an apparent trend of centralization. Considering the same treatment period, pure water treatment is most effective in mechanically weakening shale M-1 both after 20 days and 30 days, followed by treatment with supercritical $CO_2$–water and dry supercritical $CO_2$. For shale M-2, the mechanical properties after 30-day exposure to dry supercritical $CO_2$ did differ inapparently compared with the data of 10-day exposure. Throughout a longer supercritical $CO_2$–water treatment, the elastic moduli and hardness of the shale M-2 have obviously reduced. The measured hardness after 20-day supercritical $CO_2$ treatment and pure water treatment is particularly narrow distributed and exhibits a small value. This could happen when the indenter tip lands on an organic-rich surface area, where compliant compounds form a substrate. In this case, the indenter undergoes extra penetration and increases the recorded displacement, which leads to low values in hardness. From the results of shale I-1, there is no consistent tendency in mechanical evolutions with respect to the experimental condition or treatment durations. On the surface of I-1, two kinds of regions are observed: (a) a fairly homogeneous organic-rich area with spotty inorganic particles, and (b) a heterogeneous region containing abundant inorganic particles (mainly silicates), as shown in Fig. 6c. The mechanical properties obtained in the two regions can be distinctive due to the mineralogy differences. As shown in Fig. 14, the inorganic particle region has significantly stronger mechanical properties than the organic-rich region, as well as a significant heterogeneity in values. According to our results, the nano-mechanical characterization in shale highly depends on the localized mineralogy, making it difficult to compare the probing results obtained at different sites. Therefore, the general statistical analysis by grid nanoindentation gives limited information on the quantitative measurement of shale samples. It is necessary to distinguish the specific minerals in the nanoindentation tests for further shale property assessment.

### 3.3.3 Deformation During the Loading and Holding Process

Figure 15 illustrates the load–displacement curve in the nanoindentation test. The displacement is the depth that the indenter tip penetrated the sample surface, which also represents the deformation of the sample. At the end of the holding step, the sample exhibits its maximum deformation. Statistically, the mean value of maximum deformations of M-1, M-2, and I-1 shale samples before and after fluid treatment are displayed in Fig. 16. At initial, the maximum deformations of shale M-1, M-2, and I-1 are 476.1 nm, 634.9 nm, and 1047.5 nm, respectively. The more the deformation, the softer the sample is. After exposure to supercritical $CO_2$,

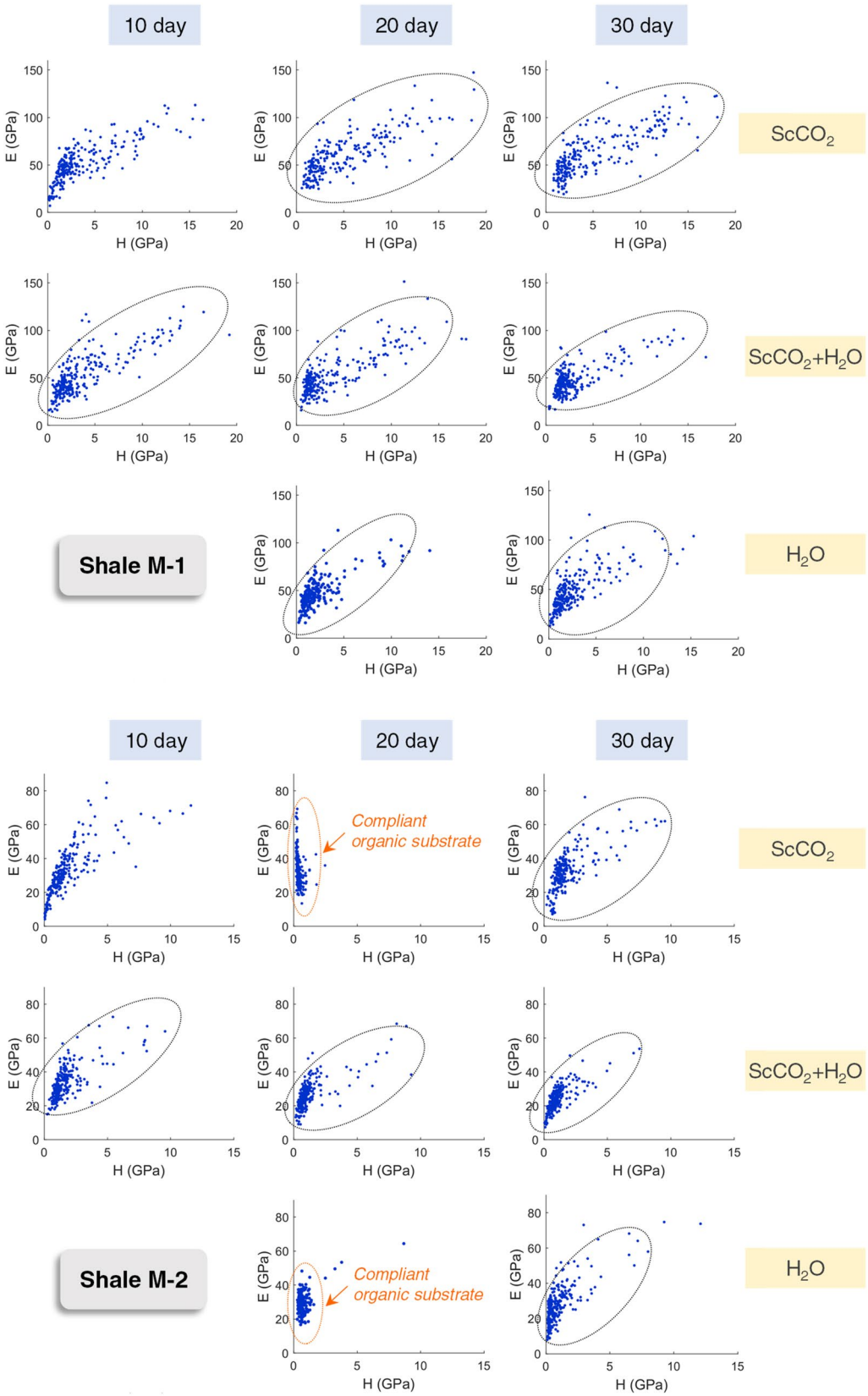



**Fig. 14** Evolution of probed mechanical properties of shale samples after treatment

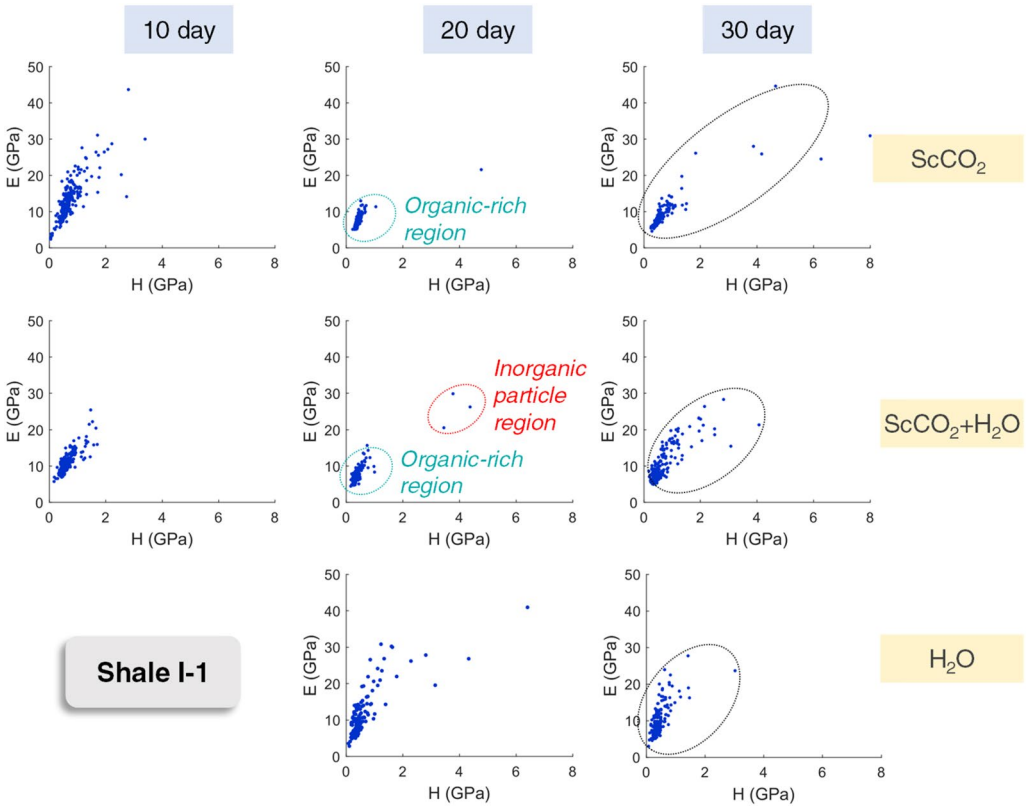



**Fig. 14** (continued)

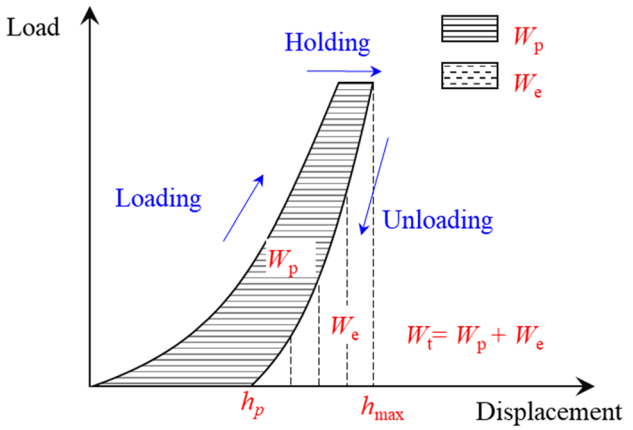



**Fig. 15** Schematic of the load–displacement curve and associated energy induced in the indentation process

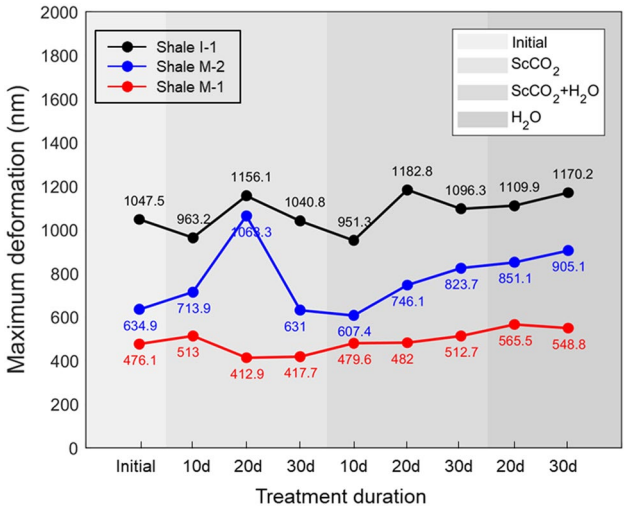



**Fig. 16** Evolution of mean maximum deformation of shale samples after supercritical $CO_2$ and water treatment

the maximum deformations of shale M-1, M-2, and I-1 are 513.0 nm/412.9 nm/417.7 nm, 713.9 nm/1063.9 nm/631 nm, and 963.2 nm/1156.1 nm/1040.8 nm, at 10/20/30 days, accordingly. After treatment with supercritical $CO_2$–water for 10/20/30 days, the maximum deformations of shale M-1, M-2, and I-1 are 479.6 nm/482.0 nm/512.7 nm, 607.4 nm/746.1 nm/823.7 nm, 951.3 nm/1182.8 nm/106.3 nm, respectively. After 30 days of pure water treatment, the maximum deformations of shale M-1, M-2, and I-1 are 548.8 nm, 905.1 nm, and 1170.2 nm. The maximum deformations of the three shale samples demonstrate an increasing trend after supercritical

$CO_2$–water treatment and pure water treatment, indicating a deterioration of the resistance to load. The maximum deformations grew by 7.7% (30 days), 29.7% (30 days), and 12.9% (20 days) for M-1, M-2, and I-1 shale, respectively, after treatment with supercritical $CO_2$–water, while grew by 18.7% (20 days), 42.6% (30 days), and 11.7% (30 days) after treatment with pure water. The maximum deformation of shale M-2 increases most significantly during the fluid treatment, while shale M-1 has the smallest change. This indicates that the fluid treatment has the most obvious effect on reducing the deformation resistance of M-2 shale, while shale M-1 may be capable of resistance to deformation due to local loads resulting from fluid erosion in the subsurface.

## 3.4 Mechanical Properties of Mineral Constituents After Treatments

### 3.4.1 Elastic Modulus and Hardness

According to the different mineralogical characteristics of the three shales, the mechanical properties of several representative mineral phases can be extracted from vast nanoindentation data. Many quartz grains are identified in the M-1 sample through EDS analysis. The large particle sizes and clear boundaries of these quartz grains make it easy to screen out the residual indents that fully land on them via SEM images. The nanoindentation data correspond

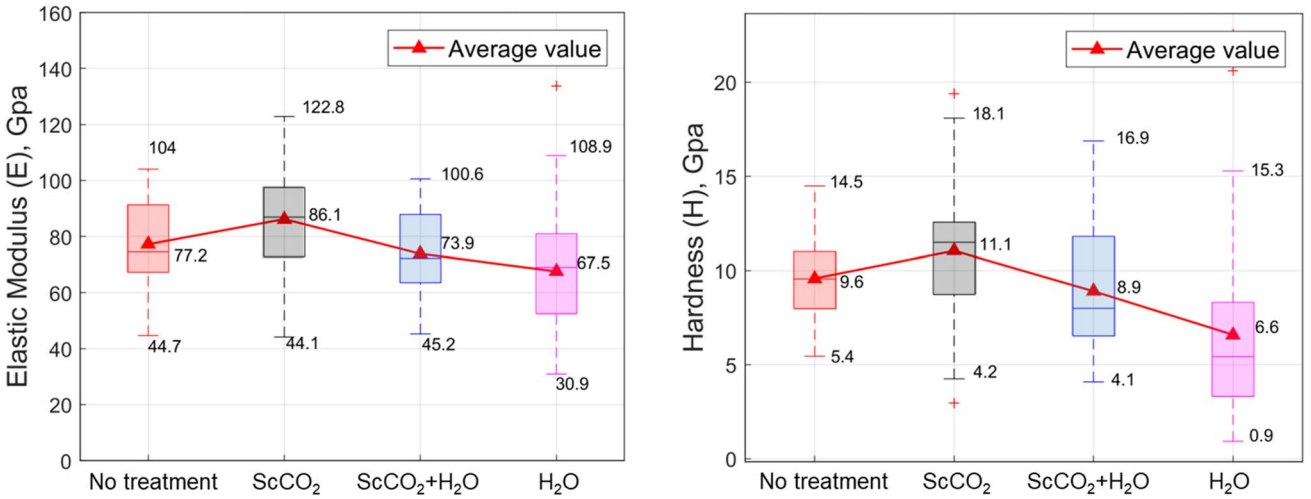



**Fig. 17** Elastic moduli and hardness of quartz in shale M-1 before and after exposure to $CO_2$ and water. The top and bottom lines indicate the maximum and minimum value, and the top edge, middle line, and bottom edge of the box represent the 25th, 50th, and 75th percentiles of the data

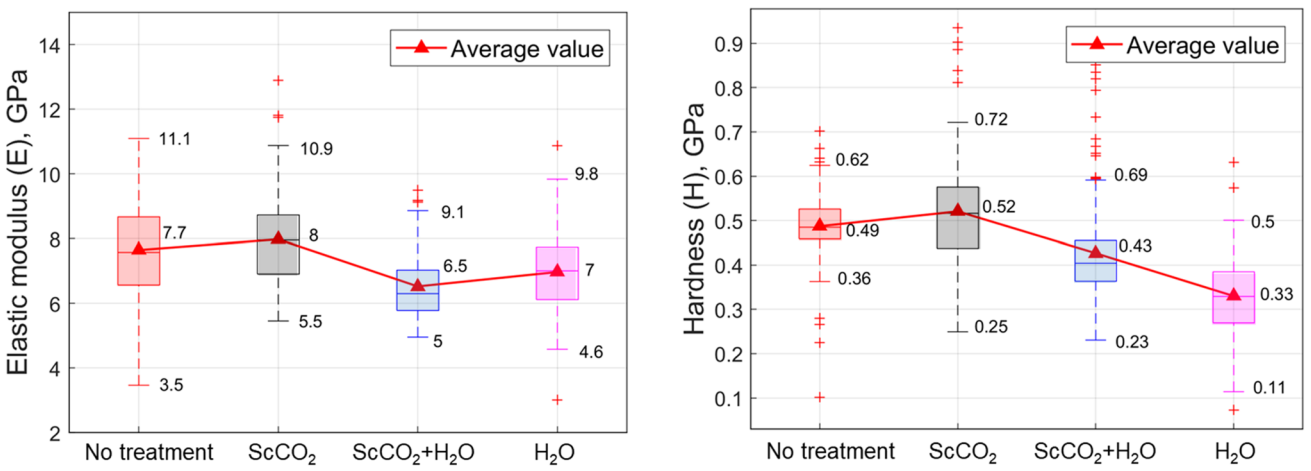



**Fig. 18** Elastic moduli and hardness of organic matter in shale I-1 before and after exposure to supercritical $CO_2$ and water

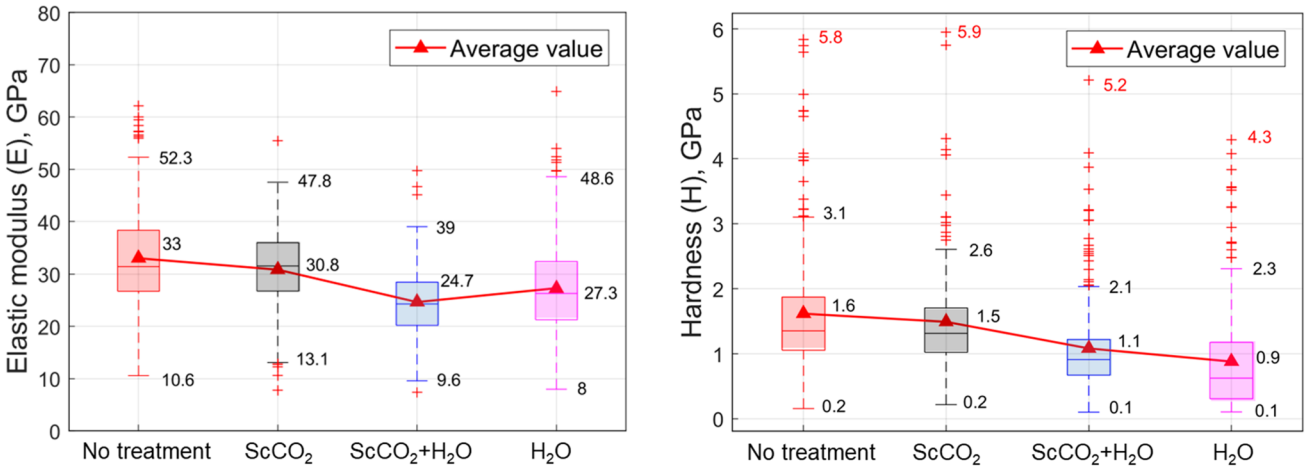



**Fig. 19** Elastic moduli and hardness of the mixed composites in shale M-2 before and after exposure to $CO_2$ and water. The mixed composites include a mixture of fine-grained clay minerals, small granular particles, and cementation

to these residual indents provide mechanical properties of quartz in shale M-1. By screening out all the quartz-related nanoindentation data in the four-round nanoindentation tests, the mechanical properties of quartz in shale M-1 before and after the fluid treatment are obtained. It is found that the quartz grains in M-2 and I-1 samples are too small and fragmented for quartz-related data extraction. However, shale I-1, which is abundantly carbonaceous, provides adequate data for tracing the mechanical properties of organic matter before and after treatments. In shale M-2, the contents of the extremely hard or soft minerals make up relatively trivial fractions. After removing coarse-grained minerals and homogenous organic matter from the data, we obtained the properties of shale composites containing fine-grained clay minerals, diminutive granular particles, and cement between the minerals. Based on this composition-based data analysis, the indentation moduli and hardness of quartz, organic matter, and mixed composite are acquired from M-1, I-1, and M-2 shale, respectively, as shown in Figs. 17, 18, and 19.

In Fig. 17, the mechanical properties of quartz are presented before and after 30-day fluid treatments. The initial mean elastic modulus of quartz in shale M-1 is 77.2 GPa. Upon exposure to supercritical $CO_2$–water for 30 days, the elastic modulus decreases slightly by 4.3%, whereas the reduction is more pronounced by 12.6% under pure water-only condition. The hardness of quartz decreases by 7.3% after 30-day treatment with supercritical $CO_2$–water; however, the mechanical weakening in hardness is more severe under pure water condition, with a reduction of 31.3%. The results are consistent with Wu's (Wu and Sharma 2017). The mean elastic modulus and hardness of quartz under dry supercritical $CO_2$ treatment are numerically affected by the maximum value within the group and thus increase compared with that under untreated condition. Previous studies show that the mechanical properties of shale are not affected by dry $CO_2$ exposure (Wu and Sharma 2017; Goodman et al. 2021). Furthermore, the measured elastic modulus and hardness of quartz vary enormously under pure water condition, indicating apparent mechanical or structural change might have occurred in the shale sample.

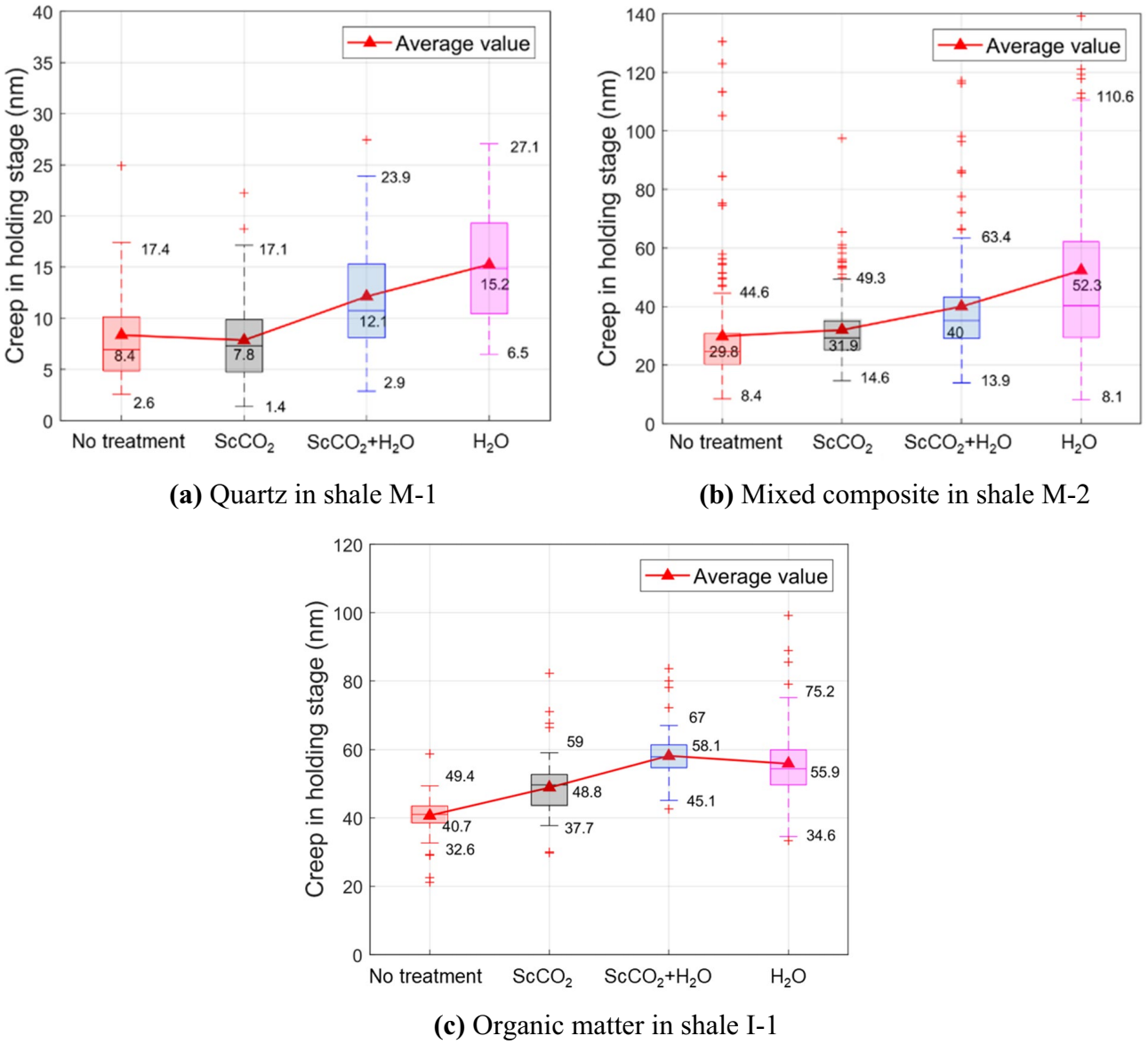
The probed mechanical properties of organic matter are extracted from the nanoindentation data of I-1 shale, in which the most surface is occupied by organic materials as demonstrated by SEM and EDS images. Figure 18 presents the mechanical properties of organic matter before and after 30-day fluid treatments. The mean elastic modulus of organic matter before treatment is 7.7 GPa, and the hardness is 0.49 GPa. After 30 days of treatment with dry supercritical $CO_2$, supercritical $CO_2$–water, and pure water, the mean elastic moduli are 8 GPa, 6.5 GPa, and 7 GPa, and the hardness are 0.52 GPa, 0.43 GPa, and 0.33 GPa, respectively. Statistical analysis shows, that after treatment with the water-contained fluid, both elastic modulus and hardness of organic matter experience apparent reductions, in which the experimental group with supercritical $CO_2$ presents the lowest elastic moduli, while the pure water group presents the lowest hardness. The average elastic moduli decrease 15.6% and 7.7% after 30 days of exposure to supercritical $CO_2$–water and water-only conditions, respectively, while the average hardness decreased 12.2% and 32.7%. The different responses indicate that additional mechanisms may be involved in the reaction of organic matter with supercritical $CO_2$ contained aqueous. The mechanical properties of organic matter exposed to dry $CO_2$ show a slight increase which may be due

**Table 3** Mechanical reductions of the main mineral phases after 30-day fluid treatment

| | Elastic modulus (*E*) | | Hardness (*H*) | |
|---|---|---|---|---|
| | $ScCO_2$ and water (%) | Water (%) | $ScCO_2$ and water (%) | Water (%) |
| Quartz | 4.3 | 12.6 | 7.3 | 31.3 |
| Organic matter | 15.6 | 7.7 | 12.2 | 32.7 |
| Matrix | 25.2 | 17.3 | 31.3 | 43.8 |

to the presence of large outliers. The results did not suggest an apparent weakening impact of dry supercritical $CO_2$ on the elastic modulus and hardness of organic matter.

Figure 19 shows the mechanical properties of mixed composite in M-2 shale matrix, including the clay minerals, small granular particles, and cementation. Statistical analysis indicates that the mean elastic moduli of the mixed composite are 33 GPa, 30.8 GPa, 24.7 GPa, and 27.3 GPa, before treatment, after treatment with dry supercritical $CO_2$, supercritical $CO_2$–water, and pure water, respectively. The hardness are 1.6 GPa, 1.5 GPa, 1.1 GPa, and 0.9 GPa, accordingly. All experimental groups show a reduction in mechanical properties after fluid treatment. The mechanical weakening of shale matrix treated with dry supercritical $CO_2$ after 30 days is the least, with a reduction of 6.7% in elastic modulus and 6.3% in hardness. This may be caused




**(a)** Quartz in shale M-1

**(b)** Mixed composite in shale M-2

**(c)** Organic matter in shale I-1

**Fig. 20** Creep during the holding stage of the three shale samples before and after exposure to $CO_2$ and water. **a** Quartz in shale M-1; **b** Mixed composite in shale M-2; **c** Organic matter in shale I-1

by a physical reaction between $CO_2$ and the clay minerals or the numerical fluctuation owing to the outliers. Upon exposure to supercritical $CO_2$–water for 30 days, the mean elastic moduli decrease by 25.2% and by 17.3% under pure water condition, while the hardness decreases by 31.3% and 43.8%, respectively. Table 3 lists the mechanical reductions of quartz, organic matter, and mixed composite after 30-day fluid treatment.

#### 3.4.2 Creep Behavior During the Holding Stage

From a variety of angles and scales, understanding the creep behavior of shale is imperative. The time-dependent mechanical behavior of shale plays a crucial role in the prediction of reservoir subsidence associated with depletion or proppant embedment (Liu et al. 2018b). As illustrated in Fig. 15, a 0.2 s holding stage begins after loading to maximum load, where the displacement continues to increase and captures the creep behavior of the material. Figure 20 shows the creep deformations of quartz, mixed composite, and organic matter in shales during the holding stage. It is obvious that the creep deformation is relatively small on the hard mineral. The creep deformations of organic matter and mixed composite before fluid treatment are 4.8 times and 3.5 times that of quartz. This indicates that the creep deformation of shale is essentially produced in organic matter and clayey matrix. After exposure to dry supercritical $CO_2$, supercritical $CO_2$–water, and pure water for 30 days, the mean creep deformations of quartz are 0.93 times, 1.44 times, and 1.81 times of creep deformation before treatment. This could be owing to the softening of the minerals around quartz particles and the possible alteration of quartz per se. The creep deformations of mixed composite exhibit 1.07 times, 1.34 times, and 1.76 times of that before treatment after 30 days of treatment with dry supercritical $CO_2$, supercritical $CO_2$–water, and pure water. The creep mechanism in the mixed composite is more complicated, which involves clay hydration, cementation weakening, and mineral dissolutions. As for the organic matter, the creep deformations after a 30-day treatment with dry supercritical $CO_2$, supercritical $CO_2$–water, and pure water are 1.20 times, 1.42 times, and 1.37 times that of no treatment, respectively. The results show that dry supercritical $CO_2$ exposure has no apparent influence on the creep behavior of quartz and mixed composite, while it will increase the creep deformation of organic matter. Previous research indicates that the pores in organic matter are isolated in two dimensions but form a connected pore network in three dimensions (Loucks et al. 2012). Supercritical $CO_2$ molecules diffuse into the organic matter pores, which may change the structures and impact the creep behaviors. Under supercritical $CO_2$–water and pure water conditions, the creep deformation of quartz, mixed composite, and organic matter all significantly increase.

To date, water is still the main component of the commercial fracturing fluid along with chemical additions. Our experimental results demonstrate that the mechanical weakening of shale due to rock–water interaction is significant at the nanoscale and mineral levels. The resistance of shale to load deteriorates dramatically after water treatment. As a result, the decrease in shale strength is related to issues with well stability and fracture closure. Inhibitor additives have been investigated to restrain the water-sensitive damage caused by clay mineral hydrations, such as potassium chloride, nonionic surfactant, polyethylene glycols, and so on (Anderson et al. 2010; Suter et al. 2011; Shadizadeh et al. 2015). However, the probing results indicate that granular particles such as quartz can be affected by fluid treatment. Thus, more mineral-involved mechanisms should be considered when dealing with the water weakening effect. $CO_2$ is injected into reservoirs for nonaqueous fracturing fluid, recovery enhancement, and $CO_2$ sequestration. Under in situ conditions, $CO_2$ is in a supercritical state. It has a chance to dissolve in the pre-existing water in reservoirs or the water that is retained due to incomplete flowback. Our experimental results show that apparent mechanical alteration occurred at the nanoscale after supercritical $CO_2$–water treatment. The weakening effects on the mineral phases are different, which may cause incoordinate deformations in response to load and induce crack initiation. This could increase the uncertainties of the safety of $CO_2$ storage, such as the potential migration of $CO_2$ and long-term leakage. Usually in hydraulic fracturing, sieved sand or ceramic proppants are added to the fracturing fluid to maintain the openness of the fractures after flowback. Conductivity is influenced significantly by the apertures of induced fractures and natural fractures, which play a role in determining gas flow efficiency. The nanoindentation results indicate that water-based fluid injections, as well as supercritical $CO_2$ fracturing (especially when water exists), will increase the creep deformations of the mineral phases in shale, which is detrimental to the reservoir conductivity. Besides, the different nanocreep behaviors between the minerals may cause damage to proppants and break the proppants into fines. On one hand, this leads to weaker prop and lowers the fracture apertures. On the other hand, the fines can migrate along the path and stagnate in fractures or void space, which may clog the flow channels and result in extra loss of fracture conductivities. The creep deformation also changes the fracture roughness, which significantly affects the fluid flow in the fracture in some situations. The creep deformation of rock may even last to production periods. From this point, the creep behavior of shale at the nanoscale ought to be considered when conducting operational design. Otherwise, the permeability may be overestimated unintentionally, and the predictions of depletion are incorrect.

As a sedimentary rock, shales have an intrinsic anisotropy developed due to the underground temperature, stresses, and sedimentary environment. Mineralogy is an important factor in the fabric anisotropy that forms the bedding planes. Anisotropy of shale at the microscale is related to the preferred orientation of clay minerals, the alignment of crystal minerals, and the placement of organics (Mba and Prasad 2010). The elastic properties of shale are determined by the interactions between its constituents and their individual components. Thus, anisotropy is a critical factor for assessing the nanomechanical properties of shale (Bennett et al. 2015). The creep deformation of shale is also observed as directional dependent (Mighani et al. 2015). It has been found that the degree of anisotropy in shale correlates with the amount of clay and organic content in the shale, as well as its fabric (Sone and Zoback 2013). The fluid treatment may also alter the anisotropy of shale. This study considers only the micromechanical properties of shale and shale minerals vertical to the bedding planes. The effects of clay orientation and mineral anisotropy on shale micromechanical evaluation should be further investigated.

## 4 Discussion

### 4.1 Involvement of Mineral Dissolution and Precipitation with Water and $CO_2$ Treatment

Shale is a multi-scale, composite sedimentary rock consisting of quartz, feldspar, clay, carbonates, organic matter, and other rock-forming minerals. Rock properties are greatly influenced by the chemical and mechanical activity of these minerals. Figure 21 shows the stability of the common compositional minerals of shale (Loucks et al. 2012; Goodman et al. 2021). It is generally assumed that silica and pyrite are mechanically and chemically stable, while clay minerals

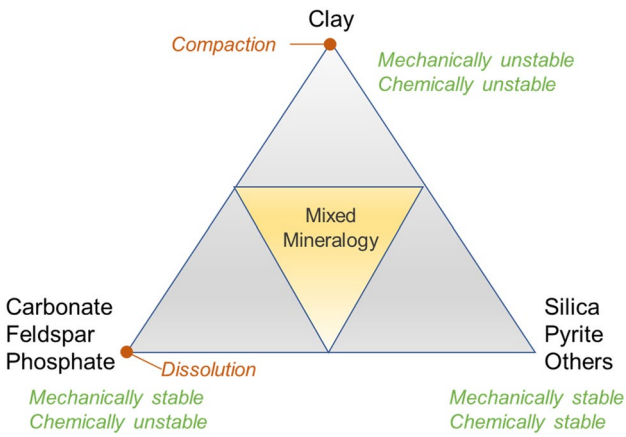



**Fig. 21** Compositional minerals of shale (Loucks et al. 2012)

are more active both chemically and mechanically. While carbonates and feldspar are relatively inactive mechanically, they are chemically unstable, with a high affinity to react or dissolve.

Quartz is a ubiquitous mineral in almost all kinds of shale and other types of rocks (Nesbitt and Young 1984). It is commonly assumed that quartz is resistant to carbonic acid and water corrosion, since it is relatively insoluble in water compared to other minerals (Schieber 2010; Lu et al. 2016a; Tian et al. 2020). However, the results in our nanoindentation tests indicate that both elastic modulus and hardness of quartz weaken after 30 days of supercritical $CO_2$ and water exposure, where the reduction in strength is more obvious with the water-only condition than in $CO_2$-rich solutions. In the investigation of the dissolution of quartz by Crundwell (Crundwell 2017), it is implied that quartz can be dissolved in water and the following equilibrium is formulated:

$$SiO_2 + 2H_2O \rightleftharpoons H_4SiO_4 \quad (1)$$

The dissolution rate is a function of the pH of the solution, temperature, and sodium ions. The dissolution rate increases as the pH value increases. That means an acid atmosphere will inhibit the dissolution of quartz and thus the dissolution rate will be larger in the absence of $CO_2$ (such as a water-only condition). In another study, amorphous $SiO_2$ formed on the surface of forsterite when it reacted with saturated supercritical $CO_2$ (Kwak et al. 2011). The dissolution of K-feldspar and kaolinite in rocks also generated $SiO_2$ (Lyu et al. 2020). Thus, quartz may be mechanically altered after $CO_2$ and water treatment due to the dissolution and precipitation of $SiO_2$. Alternatively, the surrounding microstructural may soften due to the dissolution of other minerals, affecting the probed mechanical properties of quartz. The observed in our experimental observations on the mechanical alteration of quartz raise the question of how 'inert minerals' may contribute to reservoir treatment. Further study is needed for a better understanding.

In the course of our nanoindentation experiments, we observed that organic matter experiences a significant mechanical weakening after fluid treatment. Previous studies have shown that supercritical $CO_2$ can dissolve organic matter as well as lead to a softening behavior (Jiang et al. 2016; Sheng et al. 2019). The intra-particle pores are observed in the maturation of organic matter (Loucks et al. 2012). The large pore spaces in organic matter provide sufficient contact with supercritical $CO_2$ and water, leading to its dissolution and the ultimate strength weakening. Kerogen is the most common form of immature organic matter in sedimentary rocks. According to the studies of low-maturity Marcellus shale elsewhere in the Appalachian Basin, Type II kerogen represents the main type of original organic matter (Jarboe et al. 2015). Previous studies demonstrate that the hydroxy

radicals generated by pyrite oxidation can cleave the carbon–sulfur bonds in kerogens, resulting in further precipitation of calcium sulfate that can lead to pore blockage and impede the permeability of the shale system (Goodman et al. 2021):

$$\text{S (Type II kerogen(s))} + \text{OH (aq)} \leftrightarrow SO_4^{2-}\text{ (aq)} + \text{hydrocarbons} \quad (2)$$

$$SO_4^{2-}\text{ (aq)} + Ca^{2+}\text{ (aq)} \leftrightarrow CaSO_4\text{ (s)} \quad (3)$$

In shale matrix, the involved dissolution and precipitation on the supercritical $CO_2$ and water treatments are much more complicated. Presented with water, $CO_2$ dissolves in water and forms carbonate acid solution which enters the matrix through pores and natural cracks. This causes further chemical reaction and corrosion, especially on chemically unstable minerals, such as calcite and dolomite (Wang et al. 2013; Tian et al. 2020; Goodman et al. 2021):

$$CaCO_3\text{ (calcite)} + H^+ \leftrightarrow Ca^{2+} + HCO_3^- \quad (4)$$

$$CaMg(CO_3)_2\text{ (dolomite)} + 4H^+ \leftrightarrow Ca^{2+} + Mg^{2+} + 2CO_2 + 2H_2O \quad (5)$$

Carbonic acid also attacks silicate minerals, such as albite, illite and kaolinite. Note that the K-feldspar/illite and H+ reaction is very slow but may take place on the sample surface as a slow chemical weathering process (Lyu et al. 2016, 2020; Zhang et al. 2019):

$$KAlSi_3O_8\text{ (K-Feldspar)} + 2H^+ + H_2O \leftrightarrow 2K^+ + Al_2Si_2O_5(OH)_4\text{(Kaolinite)} + 4SiO_2 \quad (6)$$

$$\text{Illite} + 11/10H^+ \leftrightarrow 23/30\text{ kaolinite} + 3/5K^+ + 1/4Mg^{2+} + 6/5\text{ quartz} + 27/20\ H_2O \quad (7)$$

$$\text{Kaolinite} + 6H^+ \leftrightarrow 5H_2O + 3Al^{3+} + 2\text{ quartz} \quad (8)$$

In addition, Fe-bearing phases, mainly including pyrite framboids and siderite, underwent oxidation in the presence of oxygen and slight dissolution after exposure to $CO_2$-saturated solution (Lu et al. 2016a; Jew et al. 2017). It was detected that Fe was released from shale to the acid solution after 96 h of interaction, followed by oxidation and precipitation reactions (Jew et al. 2017):

$$FeS_2\text{ (pyrite)} + 7O_2 + 2H_2O \leftrightarrow 2Fe^{2+} + 4SO_4^{2-} + 4H^+ \quad (9)$$

$$FeCO_3\text{ (siderite)} + 2H^+ \leftrightarrow Fe^{2+} + CO_2 + H_2O \quad (10)$$

$$2Fe^{2+}\text{ (aq)} + O_2\text{ (aq)} + 2H^+ \leftrightarrow 2Fe^{3+} + 2OH\text{ (aq)} \quad (11)$$

The above chemical reactions could occur in the $CO_2$–water–shale interaction at different time and spatial scales. On the time scale of 30 days, some chemical processes are dominant, such as (4), (8), (9), and (10). For a longer time frame, the reactions with low rates are not negligible, and their chemical behaviors affect the mechanical strength of shale in the long term.

### 4.2 The Role of $CO_2$ in the Reaction with Shale

According to our experiments, no evidence was found that dry supercritical $CO_2$ impedes the mechanical strength of shale. This is consistent with the experiment of 2021 Angela, which shows that the exposure of hale samples from different basins to dry $CO_2$ does not appear to alter the mechanical strength (Goodman et al. 2021). Kutchko et al. (2020) supported the results by reporting that Marcellus shale did not show observable alteration after exposure to $CO_2$ in the absence of water. Supercritical $CO_2$ has a high diffusion coefficient, so it can easily access any place larger than $CO_2$ molecular. In the absence of water, the reactivity of dry $CO_2$ and shale is mainly physical adsorption in organic matter or clay minerals (Kutchko et al. 2020). The fractures and pores that are accessible for $CO_2$ provide sorption sites for gas holding. Adsorption of $CO_2$ leads to volumetric expansion of shale matrix, which in turn reduces the pore spaces and fracture apertures. Physical adsorption is reversible, however. With the release of bulk pressure, $CO_2$ begins to desorb, and the damage is very limited once the $CO_2$ is completely desorbed. This explains the nonobvious mechanical alteration of the shale under dry $CO_2$ exposure. In contrast, exposure to supercritical $CO_2$ with water results in a slight reduction in elastic modulus and an obvious decrease in hardness. $CO_2$ is dissolved in water and forms a carbonate acid solution, lowering the pH of the environment. The carbonic acid solution penetrates the matrix through pores and natural cracks, and further promotes the chemical reaction and corrosion, ultimately leading to the mechanical weakening of shale. Artifacts may be introduced in the sample preparation for the nanoindentation test. On one hand, oven drying can lead to shrinkage cracks in the clay-rich shale matrix. On the other hand, the grains can be pulled out in sample polishing because of the different hardness of the components (Loucks et al. 2012). Consequently, $CO_2$-saturated solutions will have deeper intrusions due to these artifacts.

Nevertheless, previous studies show that $CO_2$ exposure changes the permeability and porosity regardless of the presence of water. Angela's experiment indicates that exposure to $CO_2$-saturated fluid improves the porosity and permeability of the Utica cores due to the mineral dissolution (Goodman et al. 2021), while dry $CO_2$ leads to a trivial change. According to the results of Wu et al. (2017), the permeability

of Utica shale was dramatically reduced after exposure to supercritical $CO_2$ due to sorption swelling. However, the increase in porosity and permeability does not necessarily facilitate the flow pathway. According to the study conducted by Elkady and Kovscek (2020) on Utica and Eagle Ford cores, the newly generated pores by carbonate dissolution are not connective pores. These pores are useless for fluid flow in the shale matrix. Whether pores generated in $CO_2$ exposure contribute to the effective flow pathway depends on the type of pores, that is, inter-particle pores or intra-particle pores, which are different in connectivity (Loucks et al. 2012; Goodman et al. 2021). Furthermore, the migration of the precipitates can block the pore throat or fracture and reduce permeability. Some relatively chemically inactive minerals, such as quartz and feldspar, may be dislodged when the adjacent carbonates dissolve. This may also lead to blockage and adversely affect the conductivity of the fracture system (Wu and Sharma 2017).

### 4.3 The Role of Water in the Reaction with Shale

The probed mechanical properties of the shale samples in our study indicate that water is indispensable to mechanical alteration. The mechanical strength is not significantly affected by 30 days of dry supercritical $CO_2$ exposure, whereas the water-containing conditions, supercritical $CO_2$–water or pure water, showed a pronounced decrease in elastic moduli and hardness. The most severe hardness degeneration appeared under water-only condition. This indicates that water dominates the interfacial contact with the minerals.

Supercritical $CO_2$ dissolves in the water and forms a carbonic acid solution. While carbonic acid is a weak acid, it is still strong enough to attack the rock-forming minerals, such as silicates, carbonates, and pyrites in short- or long-term interactions. In this process, water plays a role as a necessary reactant in generating an acidic environment. The role that water plays in the interaction is closely related to its content (Kwak et al. 2011). Dry $CO_2$ fluids can extract residual water that has condensed in pores or the water in the interlayer of swelling clay minerals with a high cation-exchange capacity, and migrate it out of the microstructure (Kwak et al. 2011; Lu et al. 2016a; Goodman et al. 2021). Although the water extracted from clay interlayers or pores is not sufficient to dissolve calcite (Elkady and Kovscek 2020), etching of the calcite surface was observed after the interaction of dry $CO_2$ with a Bakken shale containing tiny pore water (Lu et al. 2016a). This indicates local chemical reactions will take place as long as water exists. When the water content is below saturation, thin water films are formed on the surface of the minerals from nanometers to about micrometers in thickness. These films thicken with the water content (Kwak et al. 2011; Kutchko et al. 2020). In this case, the dissolution and precipitation occur at specific sites, restricted by fluid transportation. When the water content is above saturation, the water film is thicker. Sufficient water can act as a catalyst, which will contribute to the carbonate reaction and mass transport widely (Kwak et al. 2011). During our study, the samples were immersed in carbonic acid solution and pure water for 30 days. The contact between shale and water was complete, leading to chemical and corrosive deterioration. As a result, the chemically unstable minerals are dissolved in the water film that is present on the shale surface. The hydration of water-sensitive clay minerals may result in stress concentration due to the non-uniform crystalline swelling of different clay minerals, and further contribute to crack initiation (Erguler and Ulusay 2009; Anderson et al. 2010). In addition, water can penetrate the interparticle space of the minerals and impair cementation. These responses potentially accelerate the reduction of elastic modulus and hardness by the conjunction effect.

## 5 Summary and Conclusion

We investigated the nanomechanical properties of three shales and tracked the mechanical evolution after supercritical $CO_2$, supercritical $CO_2$–water, and pure water treatment for up to 30 days. The elastic modulus, hardness, and deformation behavior of the shale after fluid treatment were analyzed. Combining nanoindentation with SEM–EDS–XRD, we identified three mineral phases in the shales: quartz, organic matter, and the mixed composite. The mechanical properties and creep behaviors of the mineral phases before and after fluid treatment were analyzed. Key results from this study are summarized as follows:

1. XRD–SEM–EDS analysis and nanoindentation results demonstrate that the three shales vary significantly in mineralogy and mechanical properties. Shale M-1 is abundant in silicates. M-2 shale is found to have plenty of iron-bearing minerals, while I-1 shale presents abundant organic compounds. The shales display different mechanical properties in nanoindentation tests, resulting in varying deformation in load–displacement curves. The mechanical heterogeneity is related to the mineral constituents, which can be observed in the deconvolution results.
2. Dry supercritical $CO_2$ does not obviously impair the nanomechanical properties of shale, whereas water-contained fluid shows a significant weakening effect. After 30 days of supercritical $CO_2$–water treatment, the elastic modulus of three shales reduced up to 31.6%, and the hardness reductions are up to 20.1%. After 30 days of pure water treatment, the elastic modulus and hardness of the three shales reduced up to 47.6%. The mechanical

evolution of shale is both time-dependent and mineralogy-related.

3. The deformation behaviors of the shale were altered after fluid treatments. The maximum deformations of the three shale samples demonstrated an increasing trend after supercritical $CO_2$–water treatment and pure water treatment, indicating a deterioration of the resistance to load. The maximum deformations grew by up to 7.7%, 29.7%, and 12.9% for M-1, M-2, and I-1 shales after treatment with supercritical $CO_2$–water, while they grew by up to 18.7%, 42.6%, and 11.7% after treatment with pure water.
4. The nanomechanical properties of quartz, organic matter, and mixed composites are obtained from shale M-1, I-1, and M-2, respectively. Results show that after 30-day exposure to supercritical $CO_2$–water, the elastic modulus of quartz, organic matter, and mixed composite decreased by 4.3%, 15.6%, and 25.2%, and the hardness reduced by 7.3%, 12.2%, and 31.3%, respectively. The mechanical reductions of the three mineral phases under pure water conditions were 12.6%, 7.7%, 17.3% in elastic modulus, 31.3%, 32.7%, and 43.8% in hardness.
5. The creep deformation during the holding stage of organic matter before fluid treatment is 4.8 times that of quartz, while that of mixed composite is 3.5 times that of quartz. After fluid treatment for 30 days, the mean creep deformations of quartz can be up to 1.81 times the initial creep deformation, while that of organic matter and mix composite can be up to 1.42 times and 1.76 times, respectively. The increase in creep may damage the proppant as well as aggravate its embedment, impairing the fracture conductivity.

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